

openvaf-r — Full Change Report

Every modification and its reason (original → current)

Ngspice + OpenVAF Enhancements project

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Every modification applied to the OpenVAF-Reloaded compiler in this project, with its reason. The baseline is the pristine OpenVAF-Reloaded source tree (as vendored in the project's `original/` snapshot); the current state is the tree committed in this repository under `OpenVAF-master-20260610/`. Each entry links the enhancement write-up in `enhancements_doc/` that carries the full engineering detail and the verifying example suite. The companion document `ngspice_changes_full-report.md` covers the simulator side.

Maintenance note: this report is updated whenever an enhancement touches compiler sources. The per-enhancement index at the end tells you the last change it covers.

Scope summary. ~120 modified source files across the compiler pipeline, two new files (`hir/src/elaborate.rs` — the elaboration pass — and `hir_def/src/item_tree/diagnostics.rs`), plus three classes handled wholesale: the `test_data/` fixtures and snapshots (95 files — new test inputs and snapshots regenerated whenever a feature legitimately changed descriptor contents), the `integration_tests/*/*.osdi` reference objects (regenerated at each ABI bump), and the removal of the AGPL `va-cask` submodule. The compiler builds warning-free; every change below was regression-gated by the example-suite arsenal, the integration suite, and the `VA_TEST` industry-model corpus.

1. Lexer — `openvaf/lexer/` (`lib.rs`, `cursor.rs`)

- Based integer literals (`8'hFF`, `'sb101`): the lexer had only a commented sketch; underscores crashed it. Both the sized and unsized forms now lex, requiring a digit after the base (a silent-0 trap otherwise) ([E-46](#)).
- Don't-care digits `x/X/z/Z/?` accepted in the binary/octal/hex digit runs for `casex/casez` items ([E-78](#)).
- Escaped identifiers (`\my-net!`) lexed per the LRM ([E-46](#)).
- `// comment at EOF` hang fixed: with no trailing newline the line-comment loop spun forever on the EOF character — the only EOF-unsafe loop in the lexer ([E-35](#)).

2. Tokens and grammar — `openvaf/tokens/`, `openvaf/parser/`, `veriloga.ungram`, `sourcegen/`

`tokens/src/parser/generated.rs` gains the keywords and node kinds each language feature needed, threaded through `parser/src/grammar/*` (`stmts.rs`, `expressions.rs`, `items.rs`, `items/module.rs`), the grammar source of truth `veriloga.ungram`, and the `sourcegen/` generators (`ast/src.rs`, `hir_builtins.rs`, `mir_instructions.rs` — the templates the generated token/AST/builtin/instruction tables come from):

- `do ... while` post-test loops ([E-19](#));
- `paramset/endparamset` blocks ([E-21](#));
- `{...}` concatenation and `{n{...}}` replication as real operators, with `'{...}` remaining the typed aggregate ([E-34](#));
- array-literal expressions actually parsed (they were fully scaffolded but unreachable) ([E-4](#), [E-14](#));
- vectored (bus) net declarations and bit-selects ([E-3](#)), multi-index bit-selects for N-D arrays ([E-15](#)), and the LRM name-then-range declaration order ([E-18](#));
- module instantiation items ([E-5](#)), `generate for/genvar` ([E-8](#)), and `generate if/generate case` with optional block labels (previously mis-parsed into a broken `GENERATE_FOR`; [E-67](#));
- `defparam` — a reserved word with no grammar rule ([E-58](#));
- event OR lists `@(cross(...)` or `timer(...))` via a new or keyword and looped event grammar ([E-59](#));
- `casex/casez` sharing the case rule, the keyword token carrying the flavor ([E-78](#));

- `@(initial_step)/@(final_step)` with analysis-phase lists (E-7, E-53);
- indirect branch assignment `V(x): V(y) == expr;` (E-2);
- the `<<</>>>` arithmetic-shift tokens (plus a pre-existing lexer bug fix) (E-6);
- hierarchical parameter override targets (`#(.blk.p(4))`): the extra `.segments` are consumed so the parse stays clean instead of cascading -- elaboration then rejects the block-scoped override with a targeted diagnostic (E-87);
- hierarchical branch reference tails (`.branch(a,b) / .branch(<p>)`) swallowed into the path node so the enclosing item's CST stays whole for the elaboration rewrite (a keyword in path position used to shred the item, leaving the hole scanner truncated text) (E-86);
- part-selects in instance connections (`inst (out[3:2], in)`): the bit-select bracket accepts an optional `: expr`, with the colon token in the CST distinguishing a range from multi-dimensional indexing — no new node kind (E-85);
- module-level **generate for/if/case** without the optional **generate/endgenerate** keywords: `module_items` handled a `generate ... endgenerate` region but had no bare `FOR_KW/IF_KW/CASE_KW` arm, so a keyword-less loop at module scope fell to error recovery — unexpected token `'for'`, or a silent drop of the loop when a following analog block let recovery resync. New arms parse it into the same `GENERATE_FOR/IF/CASE` nodes the nested/generate-wrapped forms produce (E-96);
- panic-free direction parsing: `port_decl/func_arg` asserted the next token was a direction (`bump_ts`), which any non-Verilog text reaching a module-head port list turned into a compiler crash; both now emit a diagnostic with one token of forced progress (E-84);
- operator precedence corrected against LRM Table 4-2: `%` bound tighter than `*/` (so `6*7%4` evaluated to 18, not 2), and `xnor` split from `xor` (E-38);
- derived-nature parents emitted the wrong node kind (`NAME_REF` where the AST wanted `Path`), leaving the entire inheritance machinery unreachable (E-39).

3. Preprocessor — **openvaf/preprocessor/** (+ **basedb** diagnostics)

- Ten compiler directives that previously hard-failed as undefined macros: ``default_discipline`, ``celldefine/`endcelldefine`, ``unconnected_drive/`nounconnected_drive`, ``timescale`, ``line`, ``pragma`, ``undefineall`, ``default_nettype` (E-6).
- ``default_transition` parsed and threaded to `transition()` lowering (E-47).
- Recursive macro expansion crashed the compiler (stack overflow, direct and mutual): the `MacroRecursion` diagnostic existed but was never emitted, and its report renderer was a `todo!()`. The guard pushes around the macro *body* expansion only — an entry-scoped guard false-positives on the legal `QUAD(x) = TWICE(TWICE(x))` nesting (E-65).

4. Syntax / AST — **openvaf/syntax/** (**ast.rs, generated/nodes.rs, expr_ext.rs, node_ext.rs, name.rs, validation.rs, error.rs**)

- Typed AST nodes for every new construct above (generated from the ungrammar).
- Literal decoding (`expr_ext.rs`): based-int parsing with size/sign handling (E-46) and the mask-aware variant returning (`value, x_mask, z_mask`) for `casex/casez` (E-78); a compiler crash on large bare-integer-shaped literals fixed by falling back to a float constant (E-4).
- String unescaping rewritten as a single pass: the sequential `str::replace` chain corrupted overlapping escapes (`\\n`) and octal `\\ddd` was missing; all consumers route through one function (E-48).
- **Name::resolve** ate the last character of escaped identifiers (the committed `std.va` snapshot had baked the bug in as `logi`) (E-46).

5. Elaboration — `openvaf/hir/src/elaborate.rs` (new file)

The compile-time flattening pass: every stage downstream of the parser is architected around exactly one flat module per artifact, so hierarchy is resolved by recursively inlining instantiated modules — alpha-renamed per instance, ports bound to the caller's nets, parameters bound to overrides — into an ordinary flat module *before* the rest of the pipeline runs. Built for module instantiation (E-5) and grown into the home of every structural feature:

- generate `for/genvar` unrolling (E-8), nested loops, anonymous blocks, generate `if/case` with elaboration- time constant folding, and the `genvar`-substitution fix (genvars in ordinary expressions were routed through identifier renaming, which re-escaped `1e3*(i+1)` into a broken escaped identifier; they now become literal-value holes) (E-67);
- implicit nets from undeclared instance connections, discipline derived from the connected port, prefixed per instance so no cross-instance shorts (E-41);
- hierarchical names `u1.u2.node` and `$root` via an instance-chain map and a hole scanner (E-49);
- **defparam** resolved through the same chain rewrite, with the LRM-mandated precedence over instance `#( )` overrides (E-58);
- net concatenation in port connections expanded bit-by-bit onto vectored ports (E-59);
- bus slicing onto instance arrays and bus ports (E-5); paramset twin-module rendering (E-21); net initializers preserved through re-rendering (E-45); escaped identifiers re-rendered correctly (E-46);
- undefined instantiation targets are a hard error — they used to be silently dropped from the rendered output, turning a typo'd module name into an invisible open circuit; discipline-named misparses (`electrical out[0:2]`; parses as an instantiation) and paramsets dropped over unresolvable targets get tailored messages (E-84);
- name-then-range net/port declarations (`input in[0:2], electrical out[0:2]`, LRM 3.6/3.7): a textual pre-pass rewrites the `<head> <name>[range]` form to the range-then-name form (Enhancement-3), reusing all bus/port machinery; the `(-after-range` instance rule keeps instance arrays untouched (E-89). Extended to multi-name lists (`input a[0:1], b[0:3], c;`), split into one range-then-name declaration per name with per-name widths; a multi-dimensional name is left untouched (unsupported in both orders) (E-91);
- parameter-dependent declaration widths (`electrical [0:N-1] out;; real w[0:N-1];`): a textual pre-pass folds a declaration range whose bounds reference a parameter to a literal range, using the module's constant-integer parameter defaults (a fixpoint resolves one default through another; `from/exclude` constraints are skipped). Only declaration ranges (with a `:`) are folded, not bit-selects; the shared constant-integer evaluator gained an optional parameter map (empty map = the E-88 literal-only behavior). The width is fixed at the default -- a structural parameter, since OSDI has one node count per module (E-67/88 decision) (E-91). A parameter that shaped a width is then frozen to a **localparam** (`freeze_width_parameters`): a multi-parameter declaration is split so only the structural names freeze, and a range constraint is dropped from a frozen name. This keeps structure and behaviour consistent under a netlist override -- without it, overriding a width parameter that also bounds a runtime loop left the frozen array size behind, a silent out-of-bounds (E-92);
- the legacy **generate** `<id> (start, end [, incr])` statement (obsolete Verilog-A 1.0 analog-block loop-unroll, LRM Annex C.4) unrolled by a textual pre-pass: the index substitutes to a literal per iteration (with bit-select bracket folding, since a bus bit-select needs a literal index), constant bounds only -- a parameter bound cannot shape the unroll (E-67 scope decision) (E-88);
- ``__FILE__`/``__LINE__` expanded by a textual pre-pass run before the other passes (preprocessor tokens are (kind, span) pairs into existing source and cannot carry synthesized literals): ``__FILE__` becomes the root file's basename (machine-portable provenance, the E-58 rule), ``__LINE__` the exact 1-based line; string/comment occurrences are skipped and replacements are inline so later line numbers stay true (E-85);
- part-select actuals sliced onto bus ports in `bind_port (base[msb:lsb]`, constant bounds; positional and named forms; width-1 slices onto scalar ports degrade to the bit-select) with the same

ascending bit-order convention as full-bus slicing (E-85);

- block-scoped parameter override rejected with a targeted message (a named instance override `#(.blk.p(4))` naming a hierarchical target is collected and bailed like the unknown-module errors; block-scoped parameters are local to their block and take their value from their initializer, which may reference the module's parameters) (E-87);
- hierarchical branch probes (E-86): the top module's absolute chain map rides into every inlined child so SIBLING bodies resolve `V(top.a1.b)/$root...` references; unnamed-branch forms expand to the flattened node pair; port-branch probes `I(inst.branch(<p>))` read a 0V ammeter synthesized at flattening (which also fixes child-declared branch `(<p>)` port branches, broken after inlining); hierarchy-bound monitor modules are omitted from standalone output; ground/net-type declarations in inlined children keep their keyword;
- **\$port_connected** resolved at flattening time to a literal `(1)/(0)` per instance — after inlining, an open port is just a synthesized local net, so the builtin used to fail validation in exactly the unconnected case it exists to detect; top-level modules keep the native OSDI connected-flag path (E-84).

6. Definition layer — **openvaf/hir_def/** (+ **openvaf/hir/** mirrors)

`body.rs/body/lower.rs/body/pretty.rs`, `expr.rs`, `item_tree*`, `nameres*`, `builtin.rs`, `data.rs`, `types.rs`, `db.rs`, plus the new `item_tree/diagnostics.rs` and the hir façade (`body.rs`, `lib.rs`, `db.rs`, `diagnostics.rs`):

- statement/expression collection for every new construct: `do...while` (E-19), `repeat` and `disable` (E-9), `event OR lists` (E-59), `concat/ replication` (E-34), `array aggregates` and `whole-array assignment` (E-14), `CaseKind + per-item don't-care masks` with a `stray-literal list` (E-78);
- N-D array machinery: `per-dimension size lists`, `multi-index bit-selects`, `per-element parameter expansion` (E-15); `declaration initializers` on (N-D) arrays via a `Var::array_index` leaf split (E-43);
- **localparam** made genuinely non-overridable (it behaved exactly like `parameter`) with derived `localparams` still tracking inputs (E-9); `paramset` lowering as twin modules whose bound params are `localparams` with `override expressions` (E-21);
- `name resolution`: `electrical ground gnd`; `ordering` (E-9), `nature-attribute access` (`net.potential.abstol` — the third “scaffolded-but-unwired resolver boundary” found) (E-45), `derived-nature resolution` (E-39);
- **builtin.rs** (the system-function registry): the `noise-table pair` (E-9), the full `$random/$dist_*/$rdist_*` family (E-10), `file I/O and string functions` (E-11), the last fallback group (`$simprobe`, `aliases`, `plusargs`) — after which `is_unsupported()` is empty (E-12), `$simparam$str` (E-25), `$realtime` (E-59), `$table_model` (E-16);
- new diagnostics file: `wrong-arity calls` produced invalid HIR and crashed downstream — now a proper diagnostic (functions and parameters both) (E-43);
- multiple `analog blocks` collect into `entry_stmts` in source order — the `as-if-concatenated LRM semantics`, by construction (E-60);
- `bus-port node ordering` (`item_tree/lower.rs`): a non-ANSI header bus port (`module m(in, y); input [0:2] in;`) had its first bit merged into the header placeholder but the remaining bits appended after later ports, scrambling OSDI terminal order whenever the bus was not the last port. A pre-scan of body port widths now expands the bus into contiguous bits in header order at placeholder-creation time (E-90).

7. Types and validation — **openvaf/hir_ty/**

`inference.rs` (+ `inference/fmt_parser.rs`), `builtin.rs` + `builtin/generated.rs` (signature tables), `lower.rs`, `types.rs`, `validation.rs`, `validation/body.rs`, `diagnostics.rs`:

- signature tables — a recurring defect source: `ANALYSIS` made variadic (E-30); `$table_model`

given shape-synthesized varargs signatures for any dimension (a generic-varargs fallthrough *truncated* multi-arity signature lists — varargs builtins with signature lists need their own inference arm) (E-40); TRANSITION's table was one argument short (3-argument calls crashed, 4-argument worked by accident) (E-47); `transition()` input typed Real, a `$dist` typo fixed (E-49); SIMPARAM_STR returned Real instead of String (E-25);

- format-string inference (`fmt_parser.rs`): terminates on every conversion character and reports it, so `%5d/%-8s/%08x` type their arguments correctly — previously only real conversions accepted flags/width (E-71);
- array inference: element-wise array case, array-literal function arguments (which previously bound nothing and silently returned 0), integer arrays typed Real, output-literal args accepted silently — all fixed (E-33); whole-array function arguments/outputs/returns (E-18, E-20, E-23); untyped function args default to Real instead of an ICE masquerading as an initializer bug (E-43);
- **Type: :base_type** infinite loop on any array-type check (it looped on `self` instead of the cursor) — hung the compiler on simple type mismatches (E-14);
- string handling: ternary over strings (SELECT + string binary ops appended to the operator tables) (E-37);
- validation: loop bodies get their own context so the analog-operator restriction reports "loops" (LRM 4.5.1), not "conditions" (E-70); call-graph cycles among analog functions are clean errors naming the cycle instead of a recursive-inliner stack overflow (E-59); `casex/casez` restrictions (stray don't-care literals, x digits under `casez`, non-integer discriminants) (E-78); domain discrete with natures rejected per LRM 3.6.2.2 (E-50); parameter defaults exempt from their own ranges (the CMC "feature disabled" idiom — stock rejected `diode_cmc`, BSIM-CMG, PSP-HV and the HiSIM family at setup) while given values stay fully validated (E-56);
- part-selects outside port connections diagnosed (the E-78 stray-list pattern: body lowering collects them, validation reports a dedicated error naming the supported connection form) (E-85);
- contributions to port branches diagnosed (`ContributeToPortFlow`, with the declaration site labeled): `I(pb) <+ ...` on a branch (`<p>`) `pb`; slipped through the write path unvalidated and panicked during lowering (E-84);
- contributions to an all-**ground** branch diagnosed (`ContributeToGround`): `V(gnd) <+ ... / V(gnd, gnd) <+ ...` reduce both endpoints to the fixed 0 reference (no unknown) and hit an `unreachable!()` in `lower_contribute_unnamed_branch` — an ICE. The write path now checks the branch's `is_gnd` nodes and reports a clean error; a real node-to-ground branch and a `V(gnd)` probe are untouched (E-97).

8. Lowering — `openvaf/hir_lower/`

`expr.rs`, `stmt.rs`, `ctx.rs`, `callbacks.rs`, `parameters.rs`, `state.rs`, `fmt.rs`, `lib.rs`:

- analog operators: `laplace_nd/np/zd/zp` via exact state-space realization (E-4) with complex pole/zero pairs per the LRM's (re, im) convention (E-31); `zi_*` via bilinear transform (E-6); `slew/transition` as a saturating tracking loop (E-6) with the LRM negative `max_neg_rate` convention honored (the sign defect made `slew` ignore its input and ramp away) (E-61) and a DC-singularity clamp via an `EnableIntegration` identity (E-47); `idtmod`'s modulo wrap moved off the reactive residual (it diverged) and an argument- index bug fixed (E-27); `idt` initial conditions survive into transient (E-28); the `idt` assert/reset forms with smooth reset dynamics (E-52); `limexp` kept stateless as a documented decision (E-13);
- **\$table_model**: 1-D piecewise-linear lowered to differentiable MIR so autodiff yields the Jacobian for free (E-16); N-D multilinear as recursive 1-D (E-17, E-40); natural cubic splines with the precomputed moment matrix (E-22);
- events: `cross/above/timer` with persistent previous-value slots (E-8); OR lists folded with a select-based bool-or (a raw `ior` ICEs const-eval) (E-59); final-step gating and phase-list AND-ing (E-53);
- variable persistence: genuine per-instance `hidden_state` storage behind a two-pass MIR build

(E-7), typed slots so integer state stopped crashing instruction selection (E-32);

- callbacks (`callbacks.rs`, `fmt.rs`): the `$display` family with the full `[flags][width][.precision]` surface and `%h/%b/%r` translation (E-71); file I/O and string formatting through a generalized `PrintDst` sink (E-11); the deterministic seed-and-salt RNG lowering (E-10); `$sparam$str` (E-25); simulation-control return flags (E-55); `$discontinuity` (E-24);
- operator fixes: unary `~` emitted arithmetic negate; the constant folder sign-extended `>>` where the runtime was unsigned — `const` and runtime disagreed (E-37);
- whole-array coercion (`expr.rs`, `stmt.rs`): inference records a coercion of a whole array as a cast on the array *expression*, but `lower_array_elems_impl` decomposes the array and lowers each element itself, so `lower_expr's needs_cast()` never saw it — the cast was silently dead, and every new array-consuming context re-inherited the same crash (an integer element reaching a float MIR op, which `const-eval` has no case for: *"invalid operation fdiv Int(1) Float(..)"*). Patched per-call-site four times — the case discriminant (E-33), `laplace_*/zi_*` integer-literal then integer array-variable coefficients, and an integer case item against a real discriminant — before being fixed at the chokepoint, which now honours the recorded cast for every consumer (E-214). The explicit `coerce_real` flag remains for consumers whose element type is fixed by the language rather than by an inferred cast (a coefficient vector is real per LRM 9.19, and inference records no cast for it);
- masked `casex/casez` comparisons (two `iands` ahead of the existing integer equality) (E-78); array function-argument writeback (E-20) and array returns via a function-named var-array (E-23);
- named port branches (`branch (<p>) pb`; — LRM 3.7.2): a flow probe of a `PortFlow`-kind branch routes through the same `CurrentKind::Port` param as a direct `I(<p>)`, so E-29's defining equation covers both spellings; probing one used to hit `unreachable!()` (E-84);
- literal `if` conditions lower only the taken branch — a dead analog operator (`transition()` under `if ((0))`), the exact shape the `$port_connected` rewrite produces) survived `const-folding` as a detached-but-interned op whose state setup read optimized-away values and aborted codegen (E-84);
- **case** fall-through block left unsealed: `max/min/abs` lower through `make_cond` to real control flow, so one in a case default arm moved the builder onto its own merge block — the seal at the end of the case landed there and the case's own fall-through block was never sealed ("block N is not sealed"). It is now sealed where it is created; the branch just emitted is its only predecessor (E-291).

9. MIR core and optimization — `openvaf/mir*`

`mir/` (`instructions.rs` + `instructions/generated.rs`, `dfg.rs` with `dfg/phis.rs` and `dfg/uses.rs`, `dominators.rs`, `flowgraph.rs` with `flowgraph/transversal.rs`, `layout.rs`, `cursor.rs`, `lib.rs`, `builder/generated.rs`), `mir_opt/` (`simplify_cfg.rs`, `simplify.rs`, `const_eval.rs`, `global_value_numbering.rs`), `mir_autodiff/` (`builder.rs`, `lib.rs`, `live_derivatives.rs`), `mir_llvm/` (`builder.rs`, `intrinsics.rs`), `mir_build/`, `mir_interpret/`:

- three pre-existing general CFG bugs, all found via event-counter verification: a dangling-reference bug in unreachable-block removal, a multi-exit post-dominance bug in the dominator-tree builder, and a block-merge bug that could corrupt which block the DAE builder treated as the exit (E-8);
- the post-dominator sink-root pathology that let dead-code elimination delete op-dependent `$fatal` calls (side-effecting callbacks under op-dependent control now stay in eval; control dependence computed by `arm-reachability`) (E-55);
- `split_tainted.rs` tolerates layout-detached branch instructions (a branch whose condition `const-folded` has no CFG effect to taint; it used to unwrap and panic) (E-84);
- `const-eval` agreement with runtime semantics for shifts and the bitwise-not fix (E-37);
- `autodiff`: noise operators process before any `ddt` (a shared `ddt` evaluated between two noise operators silently dropped the second source), and the reactive coupling of a `ddt`-shaped noise wave is registered so small-signal pruning keeps it (E-54);
- **llvm.fabs.f64** registered in the LLVM intrinsics table — it never had been, and the signed noise-

power convention needed it (E-42);

- new opcodes/instructions supporting the operator work (barrier/ integration-identity forms; MIR deliberately has no fabs, so lowering composes it) (E-47, E-52);
- **const_eval.rs** — folding must match codegen or decline: `eval_binary` evaluated $5/0$, $5\%0$ and $i32::MIN/-1$ *inside the compiler*, so a literal zero divisor killed it outright while a runtime one had always been accepted. It now returns `Option`, declines the undefined cases and out-of-range shifts, and folds $+/-/*$ with wrapping arithmetic to match the emitted code (E-286);
- **simplify_cfg.rs** — a const-folded branch did not flag its change, so the sweep that collects orphaned blocks never re-ran and a phi kept an edge naming a value reachable only through the deleted edge — a broken-SSA function the release build carried forward, the MIR verifier being a `debug_assert!` (E-287);
- **mir_llvm/intrinsics.rs** — two declarations disagreed with the calls emitted: `hypot` was declared with one parameter and called with two (E-288), and the overloaded `llvm.ctlz` was registered without its `.i32` type suffix — it backs `$c log2` (E-289). Both produced modules the LLVM verifier rejects, and both were invisible in release for the same reason.
- branch-to-jump rewrites must retire the condition's use: a `Branch` carries one value operand, a `Jump` none, so overwriting the instruction in place leaves the condition's use record naming an operand that no longer exists. `simplify_bb`'s empty-exit-block rewrite and `dead_code_aggressive`'s dead-block rewrite both did; the two in `const_fold_terminator` already zapped/detached first (E-294).

10. DAE construction — **openvaf/sim_back/**

`dae.rs` + `dae/builder.rs`, `topology*` (incl. `linearize.rs`, `small_signal_network.rs`), `noise.rs`, `context.rs`, `lib.rs`:

- implicit equations for indirect branch assignment (one new unknown
 - residual per statement) (E-2) and the `absdelay` synthetic two-node form (E-1) — with equation indices kept stable by mapping dead/collapsed unknowns to zero-contribution placeholders instead of renumbering;
- port-flow probes **`I(<port>)`**: a TODO stub returning 0 became a synthesized DAE unknown mirroring the node's KCL residual (E-29);
- probe-only branches: probing a never-contributed branch read 0 and an open circuit — a 0 V-source pass materializes them, enabling ideal ammeters and flow-only signal-flow disciplines (E-36);
- noise factors (`noise.rs`): equation-path noise was silently lost (never attached to the DAE); factors generalize to $re + j\omega \cdot im$ with no extra matrix unknowns (E-54); same-named source grouping metadata (E-42);
- `idt` reset-mode charge dynamics and conditional step bounding (E-52).
- voltage-source branches feeding internal nodes were open circuits at DC: the small-signal (`noise/ac_stim`) pruner keyed node registration on the branch's LIVE voltage unknown, so a pure $V(a, f) \leftarrow expr(V(a, f) \text{ never read})$ registered nothing and the node's conduction silently moved to the AC-only residual; any voltage-capable branch now disqualifies its nodes (E-86);
- probed **`V(x, y) <- 0`** branches were node-collapsed away, making `I(branch)` read zero; hint pairs whose branch current is a DAE unknown are suppressed (the unprobed collapse idiom is untouched) and `NodeCollapse::hint` tolerates suppressed pairs; pinned by the permanent `vsrc_internal_node` snapshot test. Behavior change: a collapsible branch whose current the model references (`MVSG_CMC`'s access resistances carry noise on `flow(d, drc)`) stays a real 0V source — electrically identical, two extra matrix rows (E-86);

- **linearize.rs** — one analog operator nested directly inside another (`ddt(ddt(x))`): an operator materialized as an implicit equation deletes its own instruction, but a later linear contribution holds its dimension values *outside* the data-flow graph where `replace_uses` cannot reach them — and with direct nesting that stored dimension IS the deleted result. Pending entries are now retargeted onto the implicit unknown the inner operator became, which is also the correct second-derivative formulation (E-293);
- **small_signal_network.rs** — pruning is best-effort: the linearity classifier and the dimension replay can disagree, and the pass then indexed a `val_map` key that was never inserted (“no entry found for key”). It now gives up on that value — resolving its placeholder so no invalid value survives — instead of crashing (E-292).

11. OSDI code generation — **openvaf/osdi/**

`lib.rs`, `metadata.rs` + `metadata/osdi_0_4.rs`, `inst_data.rs`, `eval.rs`, `load.rs`, `setup.rs`, `compilation_unit.rs`, `ndatable.rs`, `stdlib.c`, `build.rs`, reference headers:

- descriptor emission for every ABI evolution (see the ngspice report’s ABI table): the `absdelay/last_crossing` info arrays (E-1, E-6), `OsdiNode.nodeset` (E-45), the `ac_stim` source array + `load_ac_stim` (E-51), stride-2 signed noise pairs in `load_noise` (E-54) — the version tuple lives in `osdi/src/lib.rs`;
- **metadata.rs**: the `PARAM_FLAG_FIXED` parameter-descriptor flag (a free bit of the `flags` field, additive — no layout/ABI change) set on every `localparam`, so the simulator can warn when a netlist tries to set a non-settable parameter (a `localparam`, or a structural width parameter frozen by Enhancement-92) (E-93);
- **eval.rs**: final-step flag gating (E-53); simulation-control return flags (E-55); the `ac_stim` large-signal value (was an unreachable!() panic) (E-26);
- **inst_data.rs**: `absdelay` slot layout (E-1); hidden-state slots typed by the variable (`ty_f64` was hardcoded — integer state fed f64 loads into integer MIR ops and aborted instruction selection) (E-32);
- **compilation_unit.rs** (print codegen): the `%b segfault` — the binary-formatted string was remembered for `free()` but never pushed to the `snprintf` argument list, so the matching `%s` consumed a garbage pointer (E-71); the print-callback machinery with its shadowed-fun and volatile-table IPO traps (E-11);
- **stdlib.c**: the file-descriptor table behind `$fopen/$fdisplay/...` (E-11); the `splitmix64 osdi_rng_*` runtime (E-10); a `simplparam_str` loop/return bug (E-25);
- **ndatable.rs**: `ddt/idt` natures resolved through `resolve_nature_index` instead of panicking on derived natures (E-39);
- **setup.rs**: range validation with default-exemption (E-56); a 64-line abandoned unsafe experiment (`VoidAbortCallback`) deleted in the warning cleanup (E-66);
- **build.rs**: the copied-target `stdlib` compilation trap fixed (E-10);
- **inst_data.rs**: `$temperature` read as an operator ARGUMENT used the wrong struct-GEP type — the field type (`double`) instead of the instance-data struct — so LLVM computed the offset as a `flat5*sizeof(double)` rather than `offsetof(instance, temperature)`. The shipped compiler died with SIGSEGV on `ac_stim("ac", $temperature, 0)`; the same wrong type was on the operating-point-variable read path (`nth_opvar_ptr`) (E-290).

12. Driver, libraries, and infrastructure

- `openvaf-driver/src/main.rs`: hard errors now exit non-zero — the error arm printed the failure but fell through to a success exit, so every elaboration failure looked like a successful compile to shell scripts (a quirk first noticed during E-58’s work) (E-84);
- `openvaf-driver/src/crash_report.rs`, `linker/src/lib.rs`, `lib/bforest/` (`map.rs`, `pool.rs`, `set.rs`), `lib/typed_indexmap/` (`set.rs`): the zero-warning cleanup (`44 → 0` across

- 13 crates: lifetime annotations, dead code, nightly cfgs, an unused LLVM-prefix probe) (E-66);
- `basedb/` (`ast_id_map.rs`, `lints.rs`, `lib.rs`, `diagnostics/preprocessor_error.rs`, `diagnostics/syntax_error.rs`): AST-id plumbing for elaboration-created items (the `AstId::from_erased` placeholder used by array returns; E-23), preprocessor/syntax diagnostic rendering for the new errors;
- `.llvm-version` pins the LLVM 18 toolchain expectation; the `hir` and preprocessor Cargo.tomls carry the dependency additions their new code needed.

13. The test suite — `openvaf/openvaf/tests/`, `lib/mini_harness/`, `test_data/`, `integration_tests/`

The fork's own integration suite over real compact models had never run in this project: its OSDI loader was frozen at ABI 0.4, its optional VACASK legs panicked on a never-initialized submodule, and the tests hide behind `RUN_DEV_TESTS=1`. Fixed test-side only (E-68):

- `tests/load/osdi_0_4.rs` + `load/mod.rs`: loader structs synced to ABI 0.7; `mock_sim/mod.rs`: stride-2 noise convention; `mini_harness`: missing directories skip with a note instead of panicking;
 - VACASK removed entirely (`.gitmodules`, the submodule, 34 harness legs and their stale snapshots): AGPL-3.0 cannot be vendored here;
 - `test_data/` (95 files as a class): new fixtures for new features and snapshots regenerated whenever descriptor contents legitimately changed — every regeneration reviewed (the diffs are the project's own features: `flow(<port>)` unknowns, probe-only branches, implicit-equation nodes); `integration_tests/*/*.osdi` regenerated at each ABI bump.
-
- Two correctness blind spots closed with mutation-tested guards (E-295, verification-only): the full 4x4 multi-terminal conductance AND capacitance matrices (the autodiff suite was 2-terminal plus one off-diagonal, so the KCL-derived source row and the zero body row were untested), and per-instance parameter-slot readback across 13 interleaved model/instance parameters (the guard for the E-290 offset class) (E-295).

14. Per-enhancement index (compiler-touching enhancements)

E-1 *sim_back*, *osdi*

`absdelay()` synthetic-node DAE + descriptor arrays

E-2 *parser*→*hir_lower*, *sim_back*

indirect branch assignment (implicit equations)

E-3 *parser*, *hir_def*

vectored (bus) nets with bit-selects

E-4 *hir_lower*, *syntax*, *parser*

`laplace_*` state-space + array-literal parsing

E-5 *elaborate.rs* (new), *parser*

module instantiation via compile-time flattening

E-6 *preprocessor*, *parser*, *hir_lower*, *osdi*

directives, shifts, slew/transition, `zi_*`, `last_crossing`

- E-7 *hir_lower (state), osdi*
@ (initial_step) gating + variable persistence
- E-8 *parser, elaborate, hir_lower, mir/mir_opt*
generate for + events; three general CFG bug fixes
- E-9 *hir_def, hir_lower, sim_back*
noise tables; localparam/ground/strings; repeat/disable
- E-10 *hir_lower, osdi stdlib*
\$random/\$dist_* deterministic RNG
- E-11 *hir_lower, osdi*
file I/O + string functions (PrintDst)
- E-12 *hir_def, hir_lower*
last unsupported builtins as LRM fallbacks
- E-13 *hir_lower*
limexp kept stateless (documented decision)
- E-14 *hir_def, hir_ty, hir_lower*
array literals/params/dynamic indexing; base_type hang
- E-15 *hir_def, hir_ty, hir_lower*
N-D arrays
- E-16/17/22/40 *hir_ty, hir_lower*
\$table_model: 1-D, N-D multilinear, cubic spline, varargs sigs
- E-18/20/23/33 *hir_ty, hir_lower, basedb*
arrays in analog functions: in/out/return/literals + array case
- E-19 *tokens→hir_lower*
do ... while
- E-21/44 *parser, elaborate, hir_def, sim_back*
paramset + hidden system parameters
- E-24/55 *hir_lower, mir, osdi*
\$discontinuity; \$finish/\$stop/\$fatal honored
- E-25 *hir_ty, osdi stdlib*
\$simparam\$str
- E-26/51 *osdi, sim_back*
ac_stim: crash fix, then full AC-RHS injection
- E-27/28/52 *hir_lower, sim_back*
the idt family: modulo, IC, assert/reset

- E-29/36 *sim_back*
port-flow probes; probe-only branches
- E-30 *hir_ty, hir_lower*
`variadic analysis()`
- E-31 *hir_lower*
complex poles/zeros in root forms
- E-32 *osdi inst_data*
integer persistent state (crash fix)
- E-34 *tokens*→*hir_lower*
`{...}` concat + `{n{...}}` replication
- E-35 *lexer*
EOF comment hang
- E-37/38 *hir_lower, mir_opt, parser*
operator + precedence audits (`~`, `>>` fold, `%` level)
- E-39 *parser, hir_ty, osdi*
derived natures unwired at the node-kind boundary
- E-41/49/58/59 *elaborate*
implicit nets; hierarchical names; `defparam`; port concat + OR lists
- E-42/54 *sim_back noise, mir_autodiff, mir_llvm, osdi*
correlated + node-free complex noise factors
- E-43 *hir_def(+diagnostics), hir_ty*
initializers completed; arity diagnostics
- E-45 *osdi, hir_def, elaborate*
nodesets + nature-attribute access (ABI 0.5)
- E-46/48 *lexer, syntax*
escaped ids, based literals, single-pass unescaper
- E-47 *preprocessor, hir_ty, hir_lower*
``default_transition` + transition fixes
- E-50 *hir_ty validation*
domain-binding validation
- E-53 *hir_lower, osdi eval*
`@(final_step)` + phase lists
- E-56 *hir_ty, osdi setup*
parameter defaults exempt from ranges

- E-59 *tokens*→*hir_lower*, *hir_ty*
\$realtime; OR lists; recursion diagnostics
- E-61 *hir_lower*
slow sign fix (operator-args audit)
- E-65 *preprocessor*
macro-recursion guard
- E-66 *13 crates*
zero-warning cleanup (44 → 0)
- E-67 *tokens*, *parser*, *syntax*, *elaborate*
generate audit: genvar fix, nesting, if/case
- E-68 *tests*, *mini_harness*
integration suite enabled; VACASK removed
- E-70 *hir_ty* validation
loop-context diagnostics
- E-71 *hir_ty fmt*, *hir_lower fmt*, *osdi*
display-format surface + %b segfault
- E-78 *lexer*→*hir_lower*, *hir_ty*
casex/casez don't-care masks
- E-84 *parser*, *elaborate*, *hir_ty*, *hir_lower*, *mir_opt*, *driver*
LRM example sweep: 6 defect fixes (port-branch panic, parser robustness, silent undefined modules, \$port_connected on open ports, dead-op codegen, exit codes)
- E-85 *parser*, *elaborate*, *hir_def*, *hir_ty*
`\_\_FILE\_\_`/`\_\_LINE\_\_` + connection part-selects (the last two sweep findings)
- E-86 *parser*, *elaborate*, *sim_back*
hierarchical branch probes + 2 DAE fixes (V-source-to-internal open circuit, collapse-of-probed-branch)
- E-87 *parser*, *elaborate*
block-scoped parameters (validated) + clean diagnostic for the illegal `#(.blk.p())` override
- E-88 *elaborate*
legacy generate <id> (start,end) analog-block loop-unroll (textual pre-pass)
- E-89 *elaborate*
name-then-range net/port decls (textual normalize) + Annex E SPICE-primitives example library
- E-90 *hir_def*(*item_tree*)
multi-bit input bus port bit reads: pre-scan body widths so a non-last bus port's bits stay contiguous in header/terminal order

E-91 *elaborate*

multi-name name-then-range declarations + parameter-dependent declaration widths (folded from the parameter default)

E-92 *elaborate*

freeze structural (width) parameters to `localparam` (split multi-parameter decls) so a netlist override cannot desync the frozen width from behaviour

E-93 *osdi (metadata)*

flag `localparam` OSDI parameters non-settable (`PARAM_FLAG_FIXED`, additive) so ngspice can warn on a netlist override (ngspice side too)

E-96 *parser (module grammar)*

parse a module-level `generate for/if/case` without the optional `generate/endgenerate` keywords

E-97 *hir_ty (validation)*

clean diagnostic (was an ICE) for a contribution to an all-ground branch (`V(gnd) <+ ...`)

E-101 *hir_ty (builtin), mir_opt / mir_interpret / mir_llvm*

`$clog2`: accept one argument (`INT_MATH_1`, was a 2-arg signature) and compute `ceil(log2 n)` = `bit_width(n-1)` in all three backends (was `floor(log2 n)+1`, wrong on exact powers of two)

E-102 *parser (items), syntax (ungrammar + ast), hir_def (item_tree)*

name-then-range array-valued parameters (`parameter real c[0:2]`); `parameter()` accepts post-name `[range]`, `lower_param` resolves dims per name

E-103 *mir_llvm (intrinsic)*

register the `llvm.ceil.f64` intrinsic (was missing; `ceil()` of a non-constant argument crashed codegen -- `floor` was registered)

E-104 *syntax (name), hir_def (builtin), hir_ty (builtin), hir_lower (expr)*

add `$rtoi` (real->int, truncate toward zero via `ficast((x<0)?ceil:floor)`) and `$itor` (int->real) conversion builtins

E-105 *hir_lower (expr, callbacks), osdi (compilation_unit, stdlib.c)*

`$sscanf/$fscanf` honour the format conversion base -- parse the format string, dispatch integer fields to base-specific scanners (`%h/%x` hex, `%o` octal, `%b` binary)

E-106 *hir_ty (types, inference), hir (signatures), hir_lower (expr, callbacks), osdi (compilation_unit, stdlib.c)*

string relational comparison (`</<=>/>=>`) via a lexicographic `osdi_strcmp` callback (`RELATIONAL_COMPARISON` adds the STR signature; `a<op>b` lowers to `strcmp(a,b)<op>0`)

E-107 *syntax (name), hir_def (builtin), hir_ty (builtin), hir_lower (expr, callbacks), osdi (stdlib.c)*

add `$fgetc(fd)` single-character read as a `FileOp::Getc -> osdi_fgetc` (completes the file I/O family)

E-108 *syntax (name), hir_def (builtin), hir_ty (builtin), hir_lower (expr, callbacks), osdi (stdlib.c)*

add `$ungetc(c, fd)` one-character pushback as a 2-arg `FileOp::Ungetc -> osdi_ungetc` (companion to `$fgetc`)

E-109 *hir_lower (callbacks), osdi (load)*

correct noise_table (piecewise-linear in f) and noise_table_log (log-log, $P=10^{\text{lerp}(\log_{10} p, \log_{10} f)}$) interpolation to LRM 4.6.4.3/4.6.4.4 -- both were nonconformant (lin-log, and a mis-keyed log-freq input)

E-147 *hir_ty (validation/body)*

fix exponential-time compilation of nested `?:`. Found by a robustness campaign (117 production CMC models + ~50 adversarial inputs + 4000 mutation-fuzz iterations). The body validator's `validate_expr` ends by recursing into children via `walk_child_exprs`; arms that validate their own operands return first (Call, Path), but the Select (ternary) arm did NOT -- it validated `cond/then_val/else_val` via `validate_condition` and then fell through, validating `then_val/else_val` a SECOND time, so a chain of N nested `?:` was validated 2^N times. Depth ~30 (reachable via macros) hung the compiler; sample-profiling confirmed an unbounded `validate_expr`↔`validate_condition` recursion with doubling call count. Fix: add `return;` to the Select arm (the operands are already fully validated). $O(2^N) \rightarrow O(N)$: depth 40 hang → 0.09 s, depth 2000 → 1.6 s. Behaviour-preserving: all 117 models give the IDENTICAL verdict before/after (0 flips, head-to-head), BSIM4 (12.6k lines) still ~2.3 s, and *hir_ty*/*hir*/*hir_lower*/*sim_back* tests pass with no snapshot changes. Campaign also confirmed 0 panics/segfaults across 4000 fuzz iterations; remaining lower-severity findings (parser stack overflow on ~4k-32k-deep nesting, include self-recursion, huge array dimension) are documented follow-ups. Verified 7/7 (nested `cond` examples: depth 20/40/80/160 all <0.2 s, linear growth, correct piecewise value in *ngspice*)

E-148 *parser (parser, grammar/expressions), preprocessor (diagnostics, processor), basedb (diagnostics/preprocessor_error), hir_def (item_tree, item_tree/diagnostics, item_tree/lower), hir (elaborate)*

compiler hardening: closes the three lower-severity robustness findings from E-147 (pathological input → crash/hang) by turning each into a clean bounded diagnostic. (1) Parser expression-depth limit: a shared `Parser::expr_depth` counter, incremented on each `atom_expr` (recursion) and per operator in the `expr_bp` loop (operator-chain / tree depth), bounds total expression-tree depth at 1000 and recovers to the next statement boundary when exceeded — so `-----...x / x+1+1+... / ((...)) / sin(sin(...)) / 1?...:0` no longer overflow the recursive-descent parser OR a later recursive tree traversal (`expr_bp` split into a save/restore wrapper + `expr_bp_inner`; `atom_expr` into an inc/check/dec wrapper + `atom_expr_inner`; reuses the existing `err_recover/unexpected_tokens_msg` recovery). (2) `include` recursion cap: an `include_depth` counter on the preprocessor's `Processor` caps nesting at 64, emitting a new `PreprocessorDiagnostic::IncludeRecursionLimit` (rendered in *basedb*) instead of overflowing the stack on a self-including file — mirrors the E-65 macro-recursion guard. (3) Array element cap: a shared `array_elem_count` helper (product of `|msb-lsb|+1`, capped at $2^{20} \approx 1.05\text{M}$) + a new `ItemTreeDiagnostic::ArrayTooLarge`, applied at EVERY expansion site — variable arrays, parameter arrays, net/port buses, array function returns (all in *hir_def* item-tree `lower.rs`), and instance arrays in BOTH the item-tree lowering and the *hir* elaboration pass (which re-expands `dev s[0:N]()` into rendered text) — so `real x[0:100000000]` (and param/bus/instance equivalents) degrade to a single scalar with an error instead of exhausting memory. Behaviour-preserving: all 117 production CMC models give the IDENTICAL verdict before/after (0 flips, head-to-head), BSIM4 still ~2.3s; *parser*/*preprocessor*/*hir_def*/*hir_ty*/*sim_back* unit tests pass with no snapshot changes; 0 build warnings. Verified 17/17 (robustness examples: 5 deep-expression shapes + self-include + 4 huge-array kinds all error in <1s no crash/hang with expected diagnostics; valid nested-ternary-30 / parens-100 / 100-term-sum / small arrays still compile). Fully resolves the E-147 campaign's remaining findings

E-185 *mir_autodiff (builder.rs)*

autodiff audit: hypot & atan2 derivative fixes. A deep audit compared the AC small-signal conductance dI/dV (built by the *mir_autodiff* automatic differentiation that feeds AC/convergence/noise/

pz) of a battery of nonlinear laws $I=f(V)$ against the ANALYTIC $f(V)$, and caught two builtins right in VALUE (DC correct) but wrong in DERIVATIVE -- the accidental-correctness pattern, first on the compiler side. (1) `hypot(x,y)`: the autodiff rule computed $(x'+y')/(2\text{hypot})$ -- *the sqrt(x) pattern $x/(2\text{sqrt})$* misapplied to a 2-arg function -- instead of the correct $(xx'+yy')/\text{hypot}$; 28% off at $V=0.7$, $y=0.5$, only accidentally correct at $x=0.5$. Fixed by splitting `hypot` from `sqrt` in `inst_cache` (cache holds `hypot(x,y)`, not $2*\text{hypot}$) and a correct chain rule with the usual zero/one folding. (2) `atan2(x,y)`: two bugs in the cached factors feeding the shared `Pow|Atan2` chain rule $(x'*c0 + y'*c1)*c2$ -- the common factor $c2$ was (x^2+y^2) where the rule MULTIPLIES, so it needed the reciprocal $1/(x^2+y^2)$; and $c1$ (the y' factor) was $+x$ where $d/du \text{atan2} = (x'*y - y'*x)/(x^2+y^2)$ SUBTRACTS, so it needed $-x$. Wrong magnitude AND sign. Fixed: $c2 = 1/(x^2+y^2)$, $c1 = -x$. The fix is at the autodiff-rule level so it applies to resistive currents, reactive charges (verified via AC susceptance of `ddt(hypot)`), higher-order derivatives, and `ddx`. `verify_vafautodiff`: 9 checks (both arg orders, `sqrt`-form equivalence, the $V=0.5$ accidental-correctness point, `atan2` sign + `atan2(V,V)->0` cancellation, reactive susceptance) + a 15-builtin regression battery confirming the rest were already correct. Regression 150/150.

E-187 *mir_opt (simplify.rs)*

math-identity simplifier: invalid inverse-function cancellations. The algebraic simplifier (`simplify_unary_op`) rewrites $f(g(x)) \rightarrow x$ for function-inverse pairs. Six of these were applied unconditionally even though the cancellation is only valid when f is a true left inverse of g over ALL of g 's range -- so they returned the raw inner x , WRONG for ordinary finite inputs, corrupting the DC value itself (a more severe class than the derivative-only autodiff bugs). (1) $\text{asin}(\sin(x)) \neq x$ for $|x| > \pi/2$ ($\text{asin}(\sin 3) = \pi - 3 = 0.1416$). (2) $\text{acos}(\cos(x)) \neq x$ outside $[0, \pi]$ ($\text{acos}(\cos 4) = 2\pi - 4$). (3) $\text{atan}(\tan(x)) \neq x$ for $|x| > \pi/2$ -- and this is a legitimate angle-WRAP idiom the optimizer silently defeated ($\text{atan}(\tan 2) = 2 - \pi$). (4) $\text{acosh}(\cosh(x)) \neq x$ for $x < 0$ ($\cosh$ even $\rightarrow |x|$). (5,6) `sqrt(xx)` and `sqrt(x**2)` are $|x|$, not x (`sqrt((-3)^2) = 3`, returned -3). The `const-fold path (eval_unary)` evaluates each op numerically and was always correct, so constant spot-checks passed -- the bug lived only in the symbolic cancellation on runtime values. Fix: `Asin/Acos/Atan/Acosh` decline to `simplify` (return `None`); the `Sqrt` arm returns `None` (MIR has no fabs, so the `sqrt` stays and computes $|x|$). Kept the cancellations valid over the whole real line ($\tan(\text{atan})$, $\ln(\exp)$, $\text{asinh}(\sinh)$, $\text{atanh}(\tanh)$, $\sinh(\text{asinh})$, $\cosh(\text{acosh})$, $\log_{10}(\text{pow}(10, \cdot))$). Rewrites unsound only for `Inf/NaN` ($x-x > 0$, $x0 > 0$, $\sin(\text{asin}(x))$, $\exp(\ln(x))$) left as standard finite-math. Removed the now-unused `F_TWO` import. `mir/mir_opt/mir_autodiff` unit tests unchanged (13/7/19). New example `mathident_examples/verify_mathident.py`: 12 DC-value checks. Regression 151/151.

E-186 *mir_autodiff (builder.rs + lib.rs)*

autodiff audit: real-modulo (%) derivative fix. The same AC-conductance referee (E-185) caught a third builtin right in VALUE, wrong in DERIVATIVE. Verilog-A real modulo lowers to the `Frem` opcode; $x \% c = x - \text{floor}(x/c)c$ is a slope-1 sawtooth in x , so $d/dx(x \% c) = 1$ away from the wrap points -- yet `Frem` was grouped with the genuinely-constant opcodes (`floor`, `ceil`, `clog2`, `integer/bitwiseops`, `comparisons`) and forced to derivative 0 in `TWO` places : (1) *the live-derivative gate zero derivative() in lib.rs, which decides which SS A values even depend on it -- with `Frem` listed, a modulo result was declared independent of the unknown so its derivative was never return no-edge arm*). Because the gate came first the AC/Jacobian contribution of any modulo term was a dcsweep of $I = 1e - 3 * (V \text{clog2/integer ops correctly stay in the zero group. Applies everywhere the derivative is taken -- resistive } I, \text{ reactive } Q, \text{ ddx, higher orders. } \text{verify_vafautodiff extended to 14 checks (adds 5 modulo cases: two constant divisors, prefactor scaling, chain-rule } d/dV (V\%1)^2 = 2(V\%1), \text{ and floor/ceil staying 0), a separate 3-terminal probe confirming the variable-divisor branch, cross-checked via the ddx value path; the 19 mir_autodiff unit tests are unchanged and pass. Regression 150/150.}$

E-213 *syntax (lib.rs, parsing/tree_builder.rs), lexer (lib.rs), parser (grammar/paths.rs), preprocessor (parser.rs), examples/vafcrash_examples/**

crash hardening: four compiler panics. Fuzzing openvaf-r with malformed Verilog-A found four distinct panics: instead of printing a diagnostic the compiler aborted with "OpenVAF encountered a problem and has crashed!" (exit 101) and directed the user to file a bug report -- for what is often just a typo. All are reachable from ordinary source mistakes; the headline case is a module merely missing its endmodule, one of the most common Verilog-A editing errors. BUG 1 -- EOF SPAN (tree_builder.rs): a parse error at end of file built its span as TextRange::at(self.text_pos,); text_pos is already the end of the source, so adding the previous token's length produced a span running PAST the end of the file -- 6..12 for the 6-byte input "module". Mapping it back to its file then failed assert!(range.end() <= self.range.end()) in FileSpan::with_subrange (preprocessor/sourcemap.rs), crashing the compiler WHILE TRYING TO PRINT THE ERROR ("subrange 6..12 -> 6..12 must fit into the total range 0..6"). FIX: an empty range at EOF -- exactly what the adjacent expected_at field already does. Also find_ctx_range (syntax/lib.rs) maps a position through half-open [start,end) ranges that never cover the EOF position itself (pos == last_range.end() compares Less against every range); it .expect()ed and panicked, now clamps. Triggers: missing endmodule; bare module; module header then EOF; unclosed analog begin; unterminated string. BUG 2 -- REAL LITERAL (lexer/lib.rs): e/E was consumed as an exponent marker without checking an exponent follows (eat_float_exponent()'s return, which reports whether a digit was seen, was discarded), so 1e became a Float token whose text does not parse as f64 and panicked in ast::StdRealNumber::value()'s src.parse().unwrap(). FIX: an e only joins the number when a digit (optionally signed) follows; a bare 1e lexes as 1 + identifier e and surfaces as an ordinary parse error -- the approach based_literal_body already takes for a malformed 8'squark. Valid exponents (1.5e3, 2e-3, 1E6) untouched. Triggers: 1e, 1e+, 99e, 1.5e, parameter real p=2e. BUG 3 -- PREPROCESSOR EOF (preprocessor/parser.rs): previous_range() (self.full_tokens[pos], reached while building the "expected an identifier" diagnostic) and followed_by_bracket_without_space() (self.relevant_tokens[self.pos + 1u32]; "index out of bounds: the len is 2 but the index is 2") indexed the token list directly, out of bounds one past the last token -- a bare "define" ending the file. FIX: both use .get() with a sensible fallback, mirroring the .get().map_or() idiom current_range() already used. BUG 4 -- PATH() ASSERT (parser/grammar/paths.rs): path() opened with assert!(p.at_ts(PATH_SEGMENT_TS)), a precondition several callers do not check and plausible input violates: aliasparam x=5;(a literal where a parameter name belongs),aliasparam x=;, I(<1>)/I(<>)(port_flow),discipline d1=2;. FIX: drop the assert -- the next line, p.expect_ts(PATH_SEGMENT_TS), already emits "expected identifier" and returns false, so the error is reported and an empty path node completed (nature's parser guarded its own call site this way in E-39; this makes the helper safe for every caller). This is the openvaf-r counterpart to E-212 (the same campaign against ngspice) and a direct continuation of E-148: E-148 hardened against pathological DEPTH (inputs too big), E-213 covers inputs that stop too early or are malformed. No change to any accepted program -- every fix is on a path that previously aborted the compiler; generated OSDI for every existing model is identical. Verify (examples/vafcrash, 25 checks): every repro yields a clean error instead of a crash; valid code unchanged (resistor; real exponents; aliasparam bound to a real parameter; define with args). NOTE: openvaf-r installs a CUSTOM panic hook printing its own message and exiting 101 rather than dying on a signal or printing the usual Rust "panicked at" -- a crash check looking only for those two (as E-148's suite does) scores these panics as ordinary errors, so the new suite treats exit 101 and the hook message as a crash. Toolchain tests unchanged (lexer 8, preprocessor 6, basedb 9, sim_back 26, hir 15, hir_lower 4). 25 checks. Regression 173/173 (new example folder).

E-214 *hir_lower (expr.rs, stmt.rs), examples/arraycast_examples/**

whole-array type coercion: a recurring crash class, fixed at its root. An integer Value reaching a float MIR op (feq/fmul/fsub/fdiv) panics mir_opt::const_eval::eval_binary, which has no (Int, Float) case -- "invalid operation fdiv Int(1) Float(..)", exit 101, "please open an issue". (Unfolded,

the same defect instead reaches LLVM as `i32 = fadd ..., ConstantFP:f64` and aborts with "LLVM ERROR: Cannot select" -- one bug, two faces.) FOUR INSTANCES, each patched at its own call site as found: the case discriminant over an integer array (E-33, element type hardcoded real); `laplace_/zi_ integer-LITERAL` coefficients (b77266ec); `laplace_/zi_ integer ARRAY-VARIABLE` coefficients (c55812d6 -- the previous fix had reasoned "a whole-array variable reference is already real by its declaration", which is false for an integer array); and an integer case ITEM against a REAL array discriminant (8d0ab057 -- the opcode is chosen from the discriminant, Feq, but the item's elements lower to i32). ROOT CAUSE: inference DOES record the coercion (`expect( ) -> casts.insert(item, Array{Real})`), but it lands on the array EXPRESSION, and `lower_array_elems_impl` decomposes the array and lowers each element itself -- so `lower_expr's needs_cast( )` never saw it and the cast was silently DEAD. Every new array-consuming context therefore re-inherited the trap. FIX: `lower_array_elems_impl` -- the one place every whole-array consumer passes through -- now honours a recorded cast to a real target (`coerce_real |= matches!(needs_cast(expr), Some( (_, dst)) if *dst.base_type() == Type::Real)`), making inference's intent effective for all consumers instead of requiring each call site to ask. Proven in isolation: with the case-site flag disabled the structural guard alone fixes every case repro. `lower_case` additionally keeps its own `discr_op == Opcode::Feq` coercion -- choosing the opcode from the discriminant makes matching operand types that function's invariant, independent of inference's bookkeeping. NO MISCOMPILE: the integer spelling of a filter is bit-identical to the '{1.0}' spelling (worst AC diff exactly 0 dB over 13 points) and matches the analytic response to 3e-8 dB; an integer case item still selects its arm. MUTATION-TESTED: reverting the fixes makes all four repros crash (exit 101) again, so the guard is not vacuous. Also folds in two verification guards hardened during the same hunt -- `vafautodiff` (8->16 checks: cross-derivatives with both arguments live, the blind spot that hid E-185's hypot bug, plus a 44-point battery) and `operator_examples` (+14 const-vs-runtime checks: every check there used literal operands, exercising only the folder -- the side E-37's >> bug happened to be on). New `examples/arraycast_examples` (23 checks). Regression 174/174.

E-215 *hir_ty* (*builtin.rs*), *hir_lower* (*callbacks.rs*, *expr.rs*), *osdi* (*compilation_unit.rs*, *stdlib.c*), *examples/plusargs_examples/**

testplusargs / *valueplusargs* productionized. E-12 had gated these as mechanism-unavailable fallbacks (`$test` always FALSE, `$value` never wrote its output). E-215 serves them through the `simparam` channel (ngspice publishes each command-line `+name[=value]` as namespaced `simpams` -- see the ngspice report). COMPILER side: *testplusargs*("name") lowers to `simparam("testplusargs$name",0)!=0`; *valueplusargs*("name=%fmt",var) builds the key from the compile-time literal and reads by TARGET TYPE -- `$valnum` for int/real (ficast for an int target), and a NEW non-fatal `simparam_str_opt` for a string target. `simparam_str` (E-25) FATALS on an unknown name, which a `plusarg` probe must not, so `simparam_str_opt` (`stdlib.c` + the `SimParamStrOpt` callback + `compilation_unit` wiring) returns the default instead. *hir_ty*: `VALUE_PLUSARGS`'s 2nd arg became an OUTPUT target -- three signatures `Var(Integer)/Var(Real)/Var(String)` (mirroring `$random`'s seed / `$fgets`'s buffer) instead of the read-only `Val(String)` that made extraction impossible; `TEST_PLUSARGS`'s arg is now `Literal(String)`. DESIGN NOTE: reading the value through the op-dependent `simparam` channels DELIBERATELY avoids the `$sscanf` scanner, whose `scanf_begin/scan_*` thread a hidden module-global cursor with no explicit data dependency -- the `setup/eval` partitioner can hoist a bias-independent `scan_*` into instance setup while its `scanf_begin` stays in eval, so the scan reads a NULL cursor and segfaults (observed, then designed out). *valueplusargs* matches only the `name=value` form per the LRM. No OSDI ABI change; old `.osdi` unaffected. Verify (*examples/plusargs*, 12 checks both solvers). Regression 175/175 (new example folder).

E-219 *preprocessor* (*grammar.rs*), *basedb* (*diagnostics/sink.rs*), *examples/robustness_examples/**

`preprocessor` macro-argument hang + diagnostic-flood cap. Re-running the `openvaf-r` robustness campaign against the shipped binary (see the robustness report) found a FIFTH hang path the E-147/E-148 guards did not cover, at a ~2% mutation-fuzz rate. FINDING A -- INFINITE LOOP: a

backtick token followed by `( (name()` is a macro call, and the preprocessor collects its parenthesised argument list token by token in `parse_macro_call -> parse_macro_token`. The SAME non-advancing pattern exists in the `define` `PARAMETER` list loop (`parse_define`). Neither collector had a forward-progress guarantee: when a non-Macro compiler directive (`include`, `ifdef`, `endif`, `undef`, ...) appeared inside the argument list, `parse_macro_token` pushed an `UnexpectedToken` error and `RETURNED WITHOUT CONSUMING` the token, so the caller's collection loop re-examined the same directive token forever -- an unbounded diagnostics-vector growth that pins the CPU (a sample of a hung compile named the loop exactly: `process_token -> parse_macro_call -> parse_macro_token -> RawVec::grow_one -> realloc`). It is trivially reached: injecting `(` anywhere near an `include` (or any macro token) splits the directive so the rest of the file is scanned as a macro argument. (`define` did NOT trigger it -- `compiler_directive()` has no `define` arm, so it falls through to `Macro` and is consumed as a bogus macro name, which happens to advance; only the `RECOGNISED` directives hit the non-advancing branch.) A stray delimiter in a `define` parameter list that is neither an identifier, `)` nor `,` (e.g. `a / or` in a corrupted `define`) is likewise consumed by none of the loop's `expect/eat` calls, so `parse_define` spins on it. **FIX** (`grammar.rs`): guarantee forward progress -- the stray-directive branch of `parse_macro_token` consumes the offending token after reporting it (`p.bump`); and BOTH collection loops (macro-call args and `define` params) gained a backstop that records the source offset and bails with a clean `UnexpectedEof` if an iteration consumes nothing, so no non-advancing token, present or future, can spin them. No valid model puts a directive inside a macro call's parentheses, so valid input is unaffected. **FINDING B -- DIAGNOSTIC FLOOD:** with A fixed, deeply nested NON-macro garbage (e.g. `sin( x3000` in a declaration) no longer looped but still took ~40s -- it produces thousands of parse errors and `codespan_reporting` builds a full source-annotated report for EVERY one; rendering ~3000 diagnostics (each extracting and laying out its surrounding source) is the entire cost (it persists with `stderr` redirected to `/dev/null` -- it is the BUILDING, not the writing). A bounded-but-40-second rejection is still a DoS vector. **FIX** (`sink.rs`): cap the number of diagnostics the console sink `RENDERS` at 128; beyond that only the counters advance and a one-line 'further diagnostics suppressed' note prints. `summary()` still reports the true error total and the exit code is unchanged -- the standard 'too many errors' behaviour of `rustc/clang`, which does not affect any input with `<=128` diagnostics (every real model and every ordinary mistake). **RESULT:** `name(+stray include: infinite -> clean error <0.01s`; real model + injected `(':>90s-><0.05s`; `sin( x3000` in a declaration: `~40s-><1s`; a few real errors: unchanged, all shown. Continuation of E-148 (E-148 hardened the parser `expr-depth` / `include` recursion / array expansion; E-219 covers the two paths E-148 did not touch -- the preprocessor macro-arg collector and the diagnostic sink). Verify: `examples/robustness_examples` gains eight argument-collection cases -- five macro-call (`m(` then `include/ifdef/endif/undef`, and one with 4000 leading `'('`) and three `define` parameter-list (`M(a,/b)`, `M(a"b)`, `M(a;b)`) -- plus a valid macro-call-with-nested-parens regression (26/26); the production corpus (`VA_TEST/compile_all.py`) still compiles 92/92 standalone models to the identical verdict; the campaign fuzzer's ~2% hang rate (3000 iterations) drops to 0 hangs, 0 crashes. Two files (`openvaf/preprocessor/src/grammar.rs`, `openvaf/basedb/src/diagnostics/sink.rs`). No change to any valid program's output beyond the `>128`-diagnostic suppression note.

E-220 *parser* (`parser.rs`), *syntax* (`lib.rs`), *preprocessor* (`grammar.rs`, `sourcemap.rs`), *basedb* (`diagnostics.rs`), *hir_ty* (`diagnostics.rs`, `inference.rs`, `validation.rs`, `validation/body.rs`), *examples/vaflash2_examples/**

crash hardening round 2: ten compiler panics -> clean errors. A second robustness pass, continuing E-213 and E-219. Re-running the mutation fuzzer against the shipped compiler with DIVERSE compact-model seeds (BSIM/HICUM/PSP/HiSIM/VBIC/MEXTRAM/EKV) and new mutation strategies (keyword, attribute `(**)`, and bracket injection alongside byte/truncate/delimiter/

deep-nest) found ~5% of mutated inputs still CRASHED the compiler (exit 101, 'OpenVAF encountered a problem and has crashed!') rather than reporting an error. Every one is the E-213 pattern -- a panic/assert/unwrap/index a valid model never reaches but malformed input does (all caught by the E-213 panic hook, so never memory-unsafe; the bug is the crash UX + missing diagnostic). TEN root causes, all fixed: (1) parser/parser.rs -- the parser can spin in error recovery; a step counter assert!(steps<=10M,'seems stuck') panicked. It now signals EOF at the limit so parsing winds down and reports its errors. (2) basedb/diagnostics.rs -- a diagnostic built from an empty node list hit to_unified_span_list([])=>unimplemented!(); it renders label-less, anchored to the new SourceMap::root_file(). (3) preprocessor/grammar.rs -- TextRange::new(start, end) with start>end (a macro-argument/define/include span at EOF) tripped a text-size assert; five sites clamp end>=start. (4) hir_ty/diagnostics.rs + validation.rs -- 28 sites did expr_map_back[e].unwrap()/stmt_map_back[s].unwrap(); a SYNTHESIZED expression/statement has no source-map-back entry, so reporting a type error on it panicked. They resolve the span through a fallback (empty range) via one expr_range helper. (5) hir_ty/validation/body.rs -- 11 sites did expr_types[arg].unwrap_node()/unwrap_branch()/unwrap_port_flow(), which unreachable!() when a builtin (e.g. \$port_connected) or nature access gets a wrong-typed argument inference did not reject; they bail cleanly (match {Ty::X(id)=>id, _=>return}). (6) syntax/lib.rs -- to_ctx_span mapping a span across source contexts could yield start>end in TextRange::new; clamp. (7) preprocessor/sourcemap.rs -- FileSpan::with_subrange asserted the subrange fit its parent (E-213 fixed one trigger at its source; this makes the mapping itself total); clamp into the parent. (8) preprocessor/grammar.rs -- stripping include quotes with path[1..len-1] panicked on a malformed/unterminated string literal (a lone quote); saturating_sub + get make it total. (9) hir_ty/inference.rs -- resolving a call whose arguments match NO overload left the candidate list empty after the semantic-retain and candidates[0] indexed out of bounds; it falls back to the pre-filter set (mirroring the existing restore after the exact-match retain). (10) hir_ty/validation/body.rs -- the builtin-validation arms index args[0..2] assuming the call has as many arguments as the builtin requires; a call with too few (e.g. \$simparam(), \$port_connected()) indexed out of bounds. ONE entry guard skips builtin validation when args.len() < BuiltinInfo::from(call).min_args -- inference already reports the ArgCntMismatch. Method: the fuzzer classifies OK/clean-error/CRASH/HANG; each crash was triaged from its crash log to the innermost openvaf frame, grouped, and the panic read at the source; fixes were applied one cause at a time, each verified against the recorded crash corpus, then a fresh fuzz re-checked convergence (each pass exposed the next-rarest site -- a classic fuzzing tail -- until the rate hit zero). Result: the ~390-input recorded crash corpus all clean-errors; a fresh 12000-iteration fuzz on the fixed compiler is 0 crashes / 0 hangs (down from ~5%); the 92/92 standalone production models compile to the identical verdict; parser/syntax/preprocessor/basedb/hir_ty/hir/sim_back unit suites pass. Verify (examples/vafcrash2): 19 checks with hand-crafted inputs targeting each cause, MUTATION-TESTED (reverting the fixes makes the guarded inputs crash again, so the suite is not vacuous). No OSDI/ABI change; generated .osdi for every existing model is identical. Eight files. Regression 179/179 (new example folder).

E-230 *hir_def (item_tree/lower.rs), hir_ty (validation.rs), syntax (ast/expr_ext.rs), examples/vafcrash3_examples/**

crash hardening round 3: three compiler panics -> clean errors. A third robustness pass (following E-213/E-219/E-220). The production corpus recompiled (92/92 standalone, identical verdicts) plus a fresh ~19,500-iteration mutation-fuzz over diverse compact-model seeds found THREE more distinct panic root causes, all the E-213 class (caught by the panic hook, memory-safe, but a crash instead of a diagnostic). (1) hir_def/item_tree/lower.rs -- a begin : sequential block with the scope colon but a MISSING/invalid name identifier has block_scope().is_some() yet 'name = block_scope().and_then(

E-261 *mir_autodiff (builder.rs), examples/vafsqrtguard_examples/**

autodiff: guard the **sqrt()** derivative singularity. $d/dx \sqrt{x} = 1/(2*\sqrt{x})$ is +inf at

ngspice's default $x=0$ DC initial guess; openvaf-r emitted that raw $+\infty$ into the Jacobian, which produced a nan and made the operating point fail outright -- a conductor $I=K*\text{sqrt}(V)$ never found ANY DC op (dynamic/true gmin, source, and pseudo-transient stepping all died, node = nan). The tell was an internal inconsistency: Pow carried an explicit $\text{base}=0 \rightarrow \text{derivative}=0$ guard "to ensure numerical stability" while Sqrt had none, so the mathematically identical $\text{pow}(V, 0.5)$ and $V**0.5$ converged and ngspice's own behavioral B-source $\text{sqrt}(v(n))$ converged, but $\text{sqrt}(V)$ NaN-failed. (The Pow guard is itself incomplete -- a block split that only protects a bare terminal pow/sqrt ; $2.0*\text{pow}(V, 0.5)$ fails on the unmodified compiler because the downstream multiply consumes the raw derivative.) FIX (inst_cache): change the Sqrt derivative cache from $2*\text{sqrt}(x)$ to $2*\text{sqrt}(x + a)$ ($a = 1e-18$) -- the exact derivative of the smoothly regularized $\text{sqrt}(x+a)$, so the emitted derivative is $x'/(2*\text{sqrt}(x+a))$. It is FINITE at $x=0$ ($1/(2*\text{sqrt}(a)) \sim 5e8$, a large but bounded conductance $\rightarrow$ small controlled Newton steps that creep out of the singularity, exactly like the B-source); EXACT for $x>0$ (the nudge is INSIDE the root, so the perturbation is $\sim a/(2x)$, below the ULP -- no finite-bias derivative changes, higher-order derivatives and the internal $\text{sqrt}(1-x^2)$ of asin/acos/asinh/acosh/atanh included); and COMPOSABLE (being a plain value it propagates through downstream operators $K*\text{sqrt}(x)$, $1/(1+\text{sqrt}(x))$, $\exp(-\text{sqrt}(x))$, unlike the block-split guard). Two alternatives were rejected: a block-split Sqrt arm mirroring Pow (does not compose -- scaled sqrt still fails), and an additive $2*\text{sqrt}(x)+\text{delta}$ (perturbs the derivative $\sim \text{delta}/(2*\text{sqrt}(x)) \sim 1e-9$, breaking the bit-exact higher-order asin/acos autodiff unit tests) -- the $\text{sqrt}(x+a)$ form is exact to the ULP AND convergent, so NO test tolerance was loosened. $\text{pow}(x, \text{fractional})$ is unchanged (its base derivative $(y/x)*x^y$ is an $\text{inf}*0$ form in the shared Pow/Atan2 chain rule with no clean branchless regularization; $\text{sqrt}()$ is the standard spelling). Verify (examples/vafsqrtguard, both solvers): bare and strongly-scaled $K*\text{sqrt}(V)$ ($K=1,2,5$) find their true KCL op -- not nan -- and match the equivalent B-source to $\sim 1e-5$; the guarded derivative $K/(2*\text{sqrt}(V))$ is exact for $V>0$ ($\sim 1e-6$); composed $G0/(1+\text{sqrt}(V))$ converges; $\text{pow}(V, 0.5)$ now agrees with $\text{sqrt}(V)$. openvaf-r's own autodiff suite (incl. numeric third-order asin/acos exactness at $10*\text{eps}$) and the OSDI/sim_back MIR snapshots pass unchanged in value. Two source files.

E-262 *mir_autodiff(builder.rs), examples/vafsqrtguard_examples/**

autodiff: guard the **pow(x, y)** base-derivative singularity (the E-261 sqrt fix, applied to pow). For $0 < y < 1$, $d/dx \text{ pow}(x, y) = y*x^{(y-1)}$ is $+\infty$ at the $x=0$ DC initial guess -- the same singularity as $\text{sqrt}(\text{pow}(x, 0.5))$ IS $\text{sqrt}(x)$. openvaf-r builds it via the chain rule shared by Pow and Atan2: $d(\text{pow}) = (x'*\text{cache}[0] + y'*\text{cache}[1])* \text{cache}[2]$ with $\text{cache}[0]=y/x$, $\text{cache}[2]=x^y$, so the base term $x'*(y/x)*x^y$ is $+\infty*0 = \text{NaN}$ at $x=0$ (the y/x factor is $+\infty$, the x^y factor is 0) -- it NaN-poisons the Jacobian and the operating point fails. Pow carried a PARTIAL guard (a block split forcing the derivative to 0 when $\text{base}=0$), which is why a BARE terminal $\text{pow}(V, 0.5)$ converged -- but a block split cannot compose: a downstream instruction ($2*\text{pow}(V, 0.5)$, $1/(1+\text{pow}(V, 0.5))$) consumes the RAW derivative computed inside the conditional block, so scaled/combined fractional pow still NaN-failed on the unmodified compiler. FIX (inst_cache, mirroring E-261's sqrt): cache the derivative of the a-shifted $\text{pow}(x+a, y)$ ($a = 1e-18$) while the VALUE $\text{res} = x^y$ is unchanged -- $\text{cache}[0] = y/(x+a)$, $\text{cache}[1] = \ln(x+a)$, $\text{cache}[2] = (x+a)^y$. The shared chain rule then yields base term $x'*y*(x+a)^{(y-1)}$ and exponent term $y'*\ln(x+a)*(x+a)^y$, both FINITE at $x=0$ ($y*a^{(y-1)}$ is large but bounded $\rightarrow$ a controlled Newton step out of the singularity, like the sqrt guard and ngspice's B-source), EXACT for $x>0$ (the nudge is inside the power, perturbation $\sim a/x$ below the ULP), and PLAIN VALUES so they compose through downstream operators. With a composing branchless guard in place the old block-split Pow guard is removed. Atan2 -- which shares the chain-rule ARM but has its own cache -- is untouched and verified unchanged; $y \geq 1$ is a no-op (no singularity); the solution-dependent exponent term is now finite at $x=0$ too ($\ln(x+a)$ not $\ln(0)=-\infty$). $\ln(x)/1/x$ are left as-is (their VALUE, not just the derivative, is $+\infty$ at 0, so no derivative guard well-poses a model that evaluates them at 0). Verify (examples/vafsqrtguard [6]/[7], both solvers): bare and strongly-scaled $K*\text{pow}(V, Y)$ ($K=1,2,5$ at $Y=0.5, 0.3, 0.25$) find their true KCL op -- not nan -- where the unmodified compiler NaN-failed the

scaled/fractional cases; the guarded derivative $K*Y*V^{(Y-1)}$ is exact for $V>0$ ($\sim 1e-6$). openvaf-r's autodiff suite (incl. atan2 and higher-order checks) and the OSDI/sim_back MIR snapshots pass unchanged in value; the many production models using pow (BSIM/PSP/...) are bit-unchanged for $V>0$. Completes the autodiff singular-derivative work (E-261 sqrt + E-262 pow). Two source files.

E-263 *sim_back (init.rs), hir_ty (inference.rs), hir (elaborate.rs), test_data/ui/ddx.log, examples/vafcrash4_examples/**

robustness fuzzing: three compiler panics -> clean errors (the 4th such pass, following E-213/E-219/E-220/E-230). Three fuzzing strategies against the committed compiler -- byte/token mutation of the whole .va corpus, grammar-aware STRUCTURED adversarial inputs, and VALID-but-pathological modules that compile through to the backend -- surfaced one distinct panic each (all the E-213 class: panic-hook-caught, memory-safe, but a crash instead of a diagnostic). (1) *sim_back/init.rs* -- deeply nested analog operators ($I \leftarrow \text{absdelay}(\text{ddt}(\text{ddt}(\text{absdelay}(\dots))))$) produced a value tagged for the instance-setup cache whose mapping in the INIT function is a Const or has no definition at all (`ValueDef::Invalid`); `build_init_cache` assumed every cached value is a computed instruction result and called `unwrap_result()/unwrap_inst()`, and even once survived the OSDI LLVM backend read a `BuilderVal::Undef`. FIX: before allocating a cache slot, handle the non-instruction cases -- substitute a `Float/Int/Str/Bool` CONSTANT init value directly into the eval function (eval uses the init-time value; no runtime slot needed) and treat an `Invalid` mapping like the existing dead-value path (default to 0). Only a genuine instruction result takes the slot+optbarrier path, so no cache slot or codegen ever sees a constant/undefined value again. (2) *hir_ty/inference.rs* -- `ddx(V(p,n), 5)` (a `ddx` whose 2nd arg is not a potential/flow probe) crashed `hir_lower` at `value_def(unknown).unwrap_param()`. The type-checker HAD an "invalid ddx unknown" diagnostic, but its guard tested `expr` (the `ddx` call itself, ALWAYS an `Expr::Call`) instead of `unknown`, so the diagnostic was DEAD CODE and the malformed call slipped through to codegen. FIX: test `unknown` -- a non-call `unknown` (a literal, a plain variable) now raises the diagnostic and compilation stops before lowering. (This also newly (correctly) diagnoses the `ddx(1.0,1.0)/ddx(1.0,foo)` "random fuzz" cases the *ui/ddx.va* test already carried but that the dead code never rejected -- *ddx.log* updated to 8 errors.) (3) *hir/elaborate.rs* -- a malformed top-level module whose item TREE recorded an instantiation but whose parsed AST `module_items()` came back empty (parser error-recovery and item-tree construction disagreeing) crashed the E-5/E-49 hierarchy-flattening pass at `items.first().unwrap()`. FIX: an empty AST item list means nothing to flatten -- return the module text verbatim, the same no-op the pass already takes for a module with no instantiations. Behaviour-preserving for valid input: full dual-solver regression 214/214, cargo test unchanged except the intended *ddx.log* update (no MIR/OSDI snapshot changed -- no real model exercises the new paths), and a re-fuzz of the fixed compiler across all three strategies finds no surviving panics. Verify (examples/vafcrash4): a minimal reproducer per cause now compiles cleanly (nested analog ops) or clean-errors (`ddx`, malformed module), each having exited 101 on the shipped binary. Three source files + one snapshot.

E-264 *hir (elaborate.rs), osdi (lib.rs), examples/vafhang_examples/**

large instance arrays: hierarchy-flatten $O(N^2) \rightarrow O(N)$ + OSDI-codegen stack headroom for deep per-node fan-in. Two scalability/robustness defects on the same path -- compiling a module that instantiates a large array of a sub-module. (1) FLATTEN $O(N^2) \rightarrow O(N)$ (*hir/elaborate.rs*): the E-5/E-49/E-86 pass renders one textual copy of a sub-module per instance with a per-instance name prefix and rewrites hierarchical name references, running N times for `leaf u[0:N]`. Three inner steps were themselves $O(N)$, making it $O(N^2)$ -- doubling N QUADRUPLD time ($2k \approx 1.8s$, $8k \approx 30s$, $16k \approx 100s$), so a big array looked like a hang: hierarchical-name resolution (`find_instance_path_holes`) re-scanned every instance prefix per token via `.keys().any(...)`; it was called per port binding, per instance (multiplying by N again); and each per-instance scope `clone()`d the whole E-86 absolute-reference map ($O(N)$ entries). FIX (behaviour-preserving):

precompute an ANCESTOR SET once for O(1) "is this chain a hierarchical prefix?" tests (no per-token scan); a DOT-FREE EARLY-OUT (every hierarchical ref contains a ., so the common no-dot token skips resolution entirely); and share the absolute-reference map by reference -- a small `Rc<AbsPrefixes>` (map + its precomputed ancestor set) is `Rc::cloned` (O(1)) into each scope instead of deep-copied. Now O(N): 16 001 instances elaborate+compile in ~1s (was ~100s), 32 001 in ~2s; the elaborated text and every resulting `.osdi` are IDENTICAL (only faster). (2) CODEGEN STACK (`osdi/lib.rs`): when many instances contribute to the SAME node and don't collapse (distinct per-instance parameter), the residual is a chain of thousands of accumulated contributions that OpenVAF's recursive OSDI codegen + autodiff walk; that recursion ran on a rayon codegen worker whose default stack is only a few MB, so past ~8 000 contributions it aborted with thread ... has overflowed its stack / SIGABRT -- a crash on large-but-valid input. FIX: run OSDI codegen on a dedicated rayon pool built with a 256 MiB worker stack instead of the default global pool. A thread's stack size is a property of the thread, not the work -- it cannot change the emitted code (the `.osdi` is byte-equivalent modulo the linker's already-nondeterministic build id) -- it only lets the deep recursion complete; the 8 001-contribution design now compiles cleanly. Behaviour-preserving: full dual-solver regression passes and cargo test is unchanged (no MIR/OSDI snapshot moved -- the flatten fix only accelerates identical work; the stack fix is a thread property). Extreme instance counts remain bounded by the existing 1 M array/generate caps and downstream compile cost. Verify (examples/vafhang): check A (always) -- 16 001- and 32 001-instance arrays compile in ~1s/~2s within generous absolute bounds a re-introduced O(N²) would blow; check B (--slow) -- the 8 001-contribution deep-fan-in design compiles cleanly (~38s) where the pre-fix binary SIGABRT'd. Two source files.

E-265 *hir_ty* (inference.rs), examples/vaflaplace_examples/*

laplace_*/zi_* coefficient argument: compiler panic -> clean diagnostic (fifth robustness-campaign find, following E-213/-220/-230/-263). A num/den (pole/zero) argument must be a real coefficient array (LRM 9.19). `laplace_*` bypasses the generic argument checker (which would reject a bare array-variable reference before the operator's special case can accept it) and uses `infe_array_arg`, whose fallback returned the inferred type WITHOUT requiring a real value -- so a bare NET reference (`laplace_nd(1.0, 1.0, p)`), a branch, or a string was typed but never rejected, reached `hir_lower`, and panicked (`resolve_path`: "invalid HIR: path .. was not resolved", exit 101) when the coefficient elements were lowered as values. Every ordinary value context (and the `laplace INPUT` arg) already rejects a net-as-value cleanly; only the coefficient arg was different. FIX: the check goes in `infe_laplace`, NOT in the shared `infe_array_arg` -- that helper is also used by E-33 case discriminants/items and E-34 concatenations, which legitimately carry integer/string values, and forcing real there made integer case PANIC (caught in regression before shipping). `infe_laplace` now requires each num/den type to be a real/integer scalar or array (`Ty::to_value()` is None for a net/branch reference -> rejected), emitting the normal "expected real value but found .." type-mismatch; the array-literal / array-variable / scalar shapes are unchanged. A second adjacent crash from the same fuzzing -- an empty DIRECT denominator (`laplace_nd(V, 1.0, '{}')`), where the state-space realization computes `den.len()-1` and reads `den[n]`, underflowing/OOB) -- is rejected too (the operator kind is now passed to `infe_laplace` to tell a direct denominator from a `*_np/*_zp` pole list); an empty NUMERATOR (`H=0`) and an empty POLE list (denominator polynomial 1) stay legal. Behaviour-preserving: full dual-solver regression passes (incl. the `laplace/zi` filter models in the complex-pole and RF-convolution suites), cargo test UI snapshots unchanged, and a re-fuzz of 6000 random malformed `laplace_*/zi_*` calls finds no surviving panic. Verify (examples/vaflaplace, 15 checks): six malformed coefficient args (net-den/net-num/branch/string/net-in-zi_zp-roots/empty-direct-den) now clean-error; well-formed shapes (real/int array literals, scalar, array-var ref, `zi_nd`, empty numerator, empty pole list) still compile. One source file.

Enhancements not listed (57, 60, 62–64, 69, 72–77, 79–83, 94–95, 98–100) changed no compiler sources — they were validation suites, documentation, benchmark tooling, or ngspice-side work (see the [ngspice](#)

report).

E-286 *mir_opt (const_eval.rs, simplify.rs), examples/vafcodegen_examples/**

constant-folding an integer division by zero killed the compiler. `q = 5 / 0`; exited `openvaf-r` with an internal error and produced no `.osdi`, while a *runtime* zero divisor had always been accepted (the generated code just performs the division) -- an inconsistency, not a policy. CAUSE: `eval_binary`'s integer arm evaluated every opcode directly (`func.dfg.iconst(lhs / rhs)`), so with `rhs == 0` the division happens INSIDE the compiler process and the compiler is what dies; `i32 :MIN / -1` overflows the same way. The neighbouring arms had a quieter version of the same defect -- `Iadd/Isub/Imul` folded with checked arithmetic and the shifts folded by an unconstrained distance, so `2147483647 + 1` and `1 << 40` were folded in a way that does NOT match what the generated code computes (LLVM emits plain two's-complement wrapping arithmetic; a shift distance outside `0..32` is poison). This is the `const_eval == codegen` invariant: the folder must produce exactly what the runtime path would, or decline. FIX: `eval_binary` returns `Option<Value>` (the convention `eval_unary` already used) -- it declines for `div/rem` by zero, `i32 :MIN/-1`, and shift distances outside `0..32`, leaving the instruction on exactly the runtime path a non-constant divisor takes; and folds `add/sub/mul` with `wrapping_*`. The single call site in `simplify.rs` forwards the `Option`. Verify (`examples/vafcodegen`): `constfold.va` exercises all five shapes in one module and compiles, where the pre-fix compiler exited 101. `cargo test` unchanged (no MIR/OSDI snapshot moved); full dual-solver regression passes. Two source files.

E-287 *mir_opt (simplify_cfg.rs), examples/vafcodegen_examples/**

a `const-folded` branch orphaned a block, leaving a stale phi edge (broken SSA). A noise operator in an `if` `CONDITION` is zero outside noise analysis, so the optimizer proves the branch constant and folds it -- producing a function that violates SSA. CAUSE: `const_fold_terminator` rewrites the branch to a jump and calls `remove_phi_edges(dead_dst, bb)`, which fixes the phis INSIDE the newly-unreachable successor -- but the phis that actually go stale are the ones in ITS successors, which keep an edge labelled with the orphaned block, naming a value only ever available through the edge just deleted (`v22 = phi [v31, block2], [v20, block3]` where `block2` is now unreachable and `v31` is defined in a block that reaches `block3`, not `block2`). `simplify_bb` IS the pass that collects predecessor-less blocks and prunes their successors' phi edges -- but only on a `LATER` sweep, and blocks are visited in layout order so the orphan is typically already behind the cursor. The sweep never ran again because this branch of `const_fold_terminator`, unlike its `then_dst == else_dst` sibling three lines above, never set `local_changed`; with no change flagged, `iteratively_simplify_cfg` stopped with the orphan in place. Because the MIR verifier is a `debug_assert!`, the release compiler carried the malformed function forward silently. FIX: set `local_changed` after folding, so the driver sweeps again -- monotone (a folded branch becomes a jump and cannot fold again), so termination is unaffected. Verify (`examples/vafcodegen`): `orphanblock.va` compiles and simulates (`I == V/1k`); an assertions-enabled compiler now accepts it where it reported `v31` doesn't dominate use (`block11 !dom block2`). This pass runs on EVERY model, so the whole 248-model example corpus was recompiled `release-vs-release`: zero regressions. One source file, one statement.

E-288 *mir_llvm (intrinsics.rs), examples/vafcodegen_examples/**

hypot was declared with one parameter and called with two. `I(a,b) <+ hypot(V(a), V(b))` produced a module LLVM rejects ("Incorrect number of arguments passed to called function!"). CAUSE: `hypot` needs a special case in the intrinsics table (Windows spells it `_hypot`), and that case declared `&[t_f64]` while `builder.rs` emits a two-argument call. Every other binary entry -- `atan2` and `llvm.pow.f64`, both a few lines away -- is correct; `hypot` is the odd one out because it sits outside the `ifn!` macro block. Constant arguments fold before codegen, so only a runtime argument reached the bad declaration. It survived because the check that reports it (`llmod.verify_and_print()`) is a `debug_assert!` -- release builds never run the module verifier. FIX: declare it `&[t_f64, t_f64]`, matching the call and its neighbour `atan2`. Stated plainly: on arm64

macOS the malformed call still produced the RIGHT number (the extra argument lands in the register the callee reads anyway) -- this was invalid IR LLVM is licensed to miscompile under a different target, calling convention, or optimization pipeline, not a demonstrated wrong answer on this platform. Verify (examples/vafcodegen): `hypotclog2.va` uses runtime parameters and checks `hypot(3,4) == 5` exactly. One source file, one argument list.

E-289 *mir_llvm (intrinsics.rs, builder.rs), examples/vafcodegen_examples/**

llvm.ctlz was declared without its type suffix. `$clog2(n)` with a runtime argument produced invalid IR ("Intrinsic name not mangled correctly for type arguments! Should be: `llvm.ctlz.i32`"). CAUSE: `llvm.ctlz` is an OVERLOADED LLVM intrinsic -- its name must carry the type it operates on -- but it was registered and looked up under the bare name. The neighbouring overloaded entries are all spelled correctly (`llvm.pow.f64`, `llvm.sqrt.f64`, `llvm.ceil.f64`, `llvm.lround.i32.f64`); `ctlz` is the only one missing its suffix. It backs `$clog2` (which computes `bit_width(n-1)`), so every model calling `$clog2` on a non-constant argument emitted invalid IR; as with E-288 the module verifier that reports it is a `debug_assert!`, so release builds shipped it. FOUND BY replaying the committed example corpus through an assertions-enabled compiler: `clog2_examples/clog2_demo.va` -- a model that had been shipping and simulating correctly -- was rejected. FIX: register and look it up as `llvm.ctlz.i32` (both files, including the "intrinsic not found" message). Verify (examples/vafcodegen): `hypotclog2.va` checks `$clog2(100) == 7` with a runtime parameter; the pre-existing `clog2_examples` suite continues to pass and now also passes under an assertions-enabled compiler. Two source files.

E-290 *osdi (inst_data.rs), examples/vafcodegen_examples/**

\$temperature as an operator argument used the wrong struct-GEP type -- the shipped compiler SIGSEGV'd. `I(out) <+ ac_stim("ac", $temperature, 0.0)` killed `openvaf-r` with exit 139 while optimizing the model, behind the malformed IR `getelementptr inbounds double, ptr %0, i32 0, i32 5` ("Invalid indices for GEP pointer type!"). CAUSE: `LLVMBuildStructGEP2`'s first argument is the AGGREGATE being indexed -- the instance-data struct -- and the `ParamKind: :Temperature` arm passed the `FIELD` type (`cx.ty_double()`) instead. A two-index GEP is only meaningful on an aggregate, so the IR is invalid; and the offset it describes is a flat `5 * sizeof(double) = 40` bytes rather than `offsetof(instance, temperature)`. `TEMPERATURE` is field 5 and fields 0..4 are the param-given bitfield, the two Jacobian pointer arrays, the node mapping and the collapse flags -- variable-length arrays whose combined size is essentially never 40 bytes -- so even where LLVM did not crash the load landed on unrelated bytes. Every sibling gets this right (`eval_output_slot_ptr`, `temperature_loc` both pass `self.ty`). The bug was in TWO places: `load_eval_output` (the noise / `ac_stim` argument path) and `nth_opvar_ptr` (the operating-point-variable read path). Only `$temperature` read DIRECTLY as an operator argument takes this path -- a computed variable (`tk = $temperature;`) lowers to an eval-output slot and was always correct, which is why ordinary models never tripped it. As with E-288/-289 the module verifier is a `debug_assert!`, so a release build had nothing to stop it before LLVM's optimizer hit the malformed GEP. FIX: pass `inst_data.ty` at both sites. Verify (examples/vafcodegen): `tempacstim.va` compiles (pre-fix: exit 139) and through a 1 ohm load reads back the nominal 300.15 K. One source file, two call sites.

E-291 *hir_lower (stmt.rs), examples/vafcodegen_examples/**

max/min/abs in a **case** default arm left a block unsealed. `case (V(a,b)) 5.0: y = 11.0; default: y = max(3.0, 7.0); endcase` aborted the compile with "FunctionBuilder finalized, but block N is not sealed". The discriminator is sharp: `pow(2,3)` in the same arm is fine, `max` in an **ITEM** arm is fine, `max` outside a case is fine -- only a branch-lowering builtin in the **DEFAULT** arm failed. CAUSE: `max/min/abs` do not lower to a single instruction; they lower through `make_cond` to a real select with its own then/else/merge blocks (`pow` and friends emit one instruction and open no blocks). `lower_case` creates a fall-through block per case item and switches to it, but leaves it to be sealed by an `ensured_sealed()` -- at the top of the next iteration, or, for the

LAST item, after the default arm's body is lowered. `ensured_sealed()` seals whatever block the builder is currently positioned in; when the default arm's body opens blocks of its own it leaves the builder on ITS merge block, so the seal lands there (already sealed, a no-op) and the case's fall-through block is never sealed at all. FIX: seal it where it is created -- the branch just emitted is its only predecessor, so all predecessors are known and immediate sealing is correct; the later `ensured_sealed()` calls `check_is_sealed` first and become no-ops. Verify (examples/vafcodegen): `casemax.va` compiles AND still picks the right arm ($V=2 \rightarrow \text{default } \max(3,7)=7$, $V=5 \rightarrow \text{item arm 11}$); also checked for min, abs, nested combinations, an integer discriminant, and `casex`. One source file, one statement.

E-292 *sim_back (topology/small_signal_network.rs), examples/vafcodegen_examples/**

small-signal pruning indexed a key its own replay never inserted. A fuzzer input routing noise waves through `idt` into nested `laplace_nd` coefficient arrays aborted the compile with "no entry found for key". CAUSE: `prune_small_signal` moves a linear contribution into its own dimension "where possible" (its own doc comment) -- but whether it IS possible is decided by two different pieces of code: `collect_linear_contributes` classifies the contribution as linear, while the replay inside `create_dimension` is what actually BUILDS the per-dimension value and records it in `val_map`. The replay deliberately declines several shapes (an `fmul` whose BOTH operands depend on the dimension; any opcode falling through to its catch-all), so the two analyses can disagree -- and when they did, `self.val_map[&contribute]` panicked. FIX: treat the disagreement as what it is -- pruning is a best-effort optimization -- and give up on that value instead of crashing. The bail-out path must ALSO run `replace_uses(placeholder, val)`: `prune_small_signal` creates an invalid `PLACEHOLDER` value before the replay and resolves it at the end, so an early continue that skipped it would leave an invalid value in the function. The replay instructions that go unused are dead code the later DCE pass removes. Verify (examples/vafcodegen): `ssprune.va` (the reduced reproducer) compiles where the pre-fix compiler exited 101. Behaviour-preserving wherever the two analyses agree, which is every real model: the full 248-model example corpus compiles release-vs-release with zero regressions, including the noise and RF suites that exercise this pass most heavily. One source file.

E-293 *sim_back (topology/linearize.rs), examples/vafcodegen_examples/**

one analog operator nested directly inside another. `I(a,b) <+ ddt(ddt(V(a,b)))` crashed the compile -- as did three deep, or split across a variable (`x = ddt(V); I <+ ddt(x);`) -- while putting anything at all in between (`ddt(2.0*ddt(V))`) always worked. CAUSE: `build_analog_operators` materializes each classified operator; an `Evaluation::Equation` allocates an implicit unknown, calls `replace_uses(res, eq_val)` and then DELETES the operator's instruction, while an `Evaluation::Linear` adds a stored dimension value into the contribution's reactive part. Those dimension values live in the `Evaluation::Linear { contributes }` triples -- OUTSIDE the data-flow graph -- and `replace_uses` only rewrites operands held inside the DFG, so it cannot reach them. With direct nesting the stored dimension IS the inner operator's result, so once the inner operator is processed the outer one holds a value whose defining instruction has been removed (`LINEAR inst1 uses dimension=v17 (def = Result(inst0,0))` where `inst0` was just deleted); with an `fmul` in between the replay yields a fresh value (`Result(inst2)`, the multiply) and the situation never arose. Everything derived from the dangling value surfaced later as invalid argument `vN` when the instance-init function was validated. FIX: iterate the operator list by index so an operator can fix up the entries still pending behind it, and retarget any pending dimension naming this operator's result -- `eq_val`, the implicit unknown the inner operator became, is exactly what the outer operator's reactive contribution should use, so this states the correct second-derivative formulation rather than merely removing a dangling reference. The `Evaluation::Dead` arm carries the same hazard and is retargeted to `F_ZERO`; the reverse order is already safe (a `Linear` processed first inserts a genuine DFG use, which a later `Equation`'s `replace_uses` does reach). Verify (examples/vafcodegen): in AC a `ddt` is $j\omega$, so a second derivative is $(j\omega)^2 = -\omega^2$ --

E-294 *mir_opt (simplify_cfg.rs, dead_code_aggressive.rs), examples/vafcodegen_examples/**

a **Branch->Jump** rewrite left the condition in the use list. A Branch carries exactly one value operand (its condition); a Jump carries none, so every such rewrite must retire that operand's use-list entry. Two of the four sites simply overwrote the instruction (`insts[terminator] = Jump{. . }`), leaving the record linked into the condition value's list naming OPERAND 0 of an instruction that now has ZERO operands -- `use_to_value` then indexes an empty slice ("index out of bounds: the len is 0 but the index is 0"). Offenders: `simplify_bb`'s empty-exit-block rewrite (both arms), and `dead_code_aggressive`'s dead-block terminator rewrite (same defect, found by inspecting the class rather than from a failing model). The two rewrites in `const_fold_terminator` do it correctly -- one via `zap_inst`, one via `detach_operand` -- which is what made the omission legible as an inconsistency. REACHED BY a narrow shape: `$fatal` exits the analog block so its arm becomes an EMPTY EXIT block, exactly what `simplify_bb` rewrites, and the condition must be a PARAMETER compare (with a node voltage the arm is not an empty exit block and the rewrite never fires; `$finish` does not produce this shape either) -- one module in the whole corpus reached it. FIX: `zap_inst` the terminator before overwriting, at both sites. HONEST SEVERITY: the stale entry did NOT produce a wrong `.osdi` and no release build could be made to fail on it -- release never runs the MIR verifier (`debug_assert!`), and the passes that would trip over the record (`replace_uses`, `use_set_value`, whose bounds checks ARE active in release) happen never to touch that value again in any corpus model. A broken invariant with a latent release-crash hazard, not a demonstrated miscompilation -- and the last thing between the compiler and a clean assertions-enabled corpus run, the audit that produced E-286..293. A related latent case is deliberately left alone: `update_inst_uses` retires surplus use records with `truncate`, dropping them from the INSTRUCTION's list without detaching from the VALUE's -- same class, but unreachable today (every caller zaps first, and `attach_use`'s own `debug_assert` would have fired otherwise), so it is documented rather than patched speculatively. Verify (examples/vafcodegen): `staleuse.va` compiles and simulates; the authoritative check is the assertions-enabled corpus replay -- all 255 models clean, 0 latent, where `simctrl_demo.va` aborted. Release-vs-release 255/255, 0 regressions; cargo test 69/0. Two source files.

E-295 *examples/vafautodiff_examples/, examples/vafcodegen_examples/*

regression guards for the two correctness blind spots (verification-only, no source change). A correctness campaign (~150 oracle checks over parameter storage, multi-terminal Jacobian/capacitance matrices, noise, node collapsing, `$mfactor`, `$table_model`, temperature) found ZERO defects; this folds in only the two checks that were genuinely NEW coverage. NOT added because already covered: flicker-noise 1/f and the correlated-noise summation rule (`noise_examples + noisecorr_examples` assert both exactly) and 2-D `$table_model` (`mdtable` already checks the DC surface AND both partials). [1] FULL MULTI-TERMINAL MATRICES (`vafautodiff`, 16->18 checks): the suite biased only 2-terminal devices and `[cross]` read ONE off-diagonal, so the entries `openvaf` does not obtain by differentiating a contribution -- the KCL-derived source row, and the identically-zero row/column of an untouched terminal -- were untested; the new `[matrix]` checks measure all 16 entries of BOTH `dI/dV` and `dQ/dV` (separate reactive code path) on a device whose two contributions are polynomials in three distinct branch voltages at a bias where every branch voltage differs. MUTATION-TESTED: dropping the second product-rule term in `mir_autodiff (d(uv)=u'v)` makes both `[matrix]` checks FAIL (1.6e-1 / 2.2e-1) while `[cross]`, `[regression]` and `[multipoint]` all still PASS -- the new guard has UNIQUE detection power, not just breadth. [2] PARAMETER SLOTS PER INSTANCE (`vafcodegen`, 17->19 checks): E-290 was a wrong struct-GEP offset, and in a 1-2 parameter model such an error lands on the right bytes by luck -- which every prior oracle test was. `paramslots.va` interleaves 13 model+instance parameters of mixed types with distinct non-round values, mirrors each through its own `opvar`, and 3 instances across 2 model cards give 39 readbacks covering defaults, model card, instance line, instance-overrides-model and cross-instance isolation. MUTATION-TESTED: the slot index is used by BOTH writer and reader so

permuting it is a self-consistent renaming and unobservable -- only a reader/writer MISMATCH shows (which is what E-290 was); making `nth_opvar_ptr` read a different slot than eval wrote yields `mp0 = 7.0` (its neighbour `ip0`'s value) and the guard fails. HONEST LIMIT found by that mutation test: E-290 fixed two sites, and only `load_eval_output` is reachable (covered by the existing `ac_stim` check); `nth_opvar_ptr`'s `ParamKind::Temperature` arm is NOT reachable -- `ov = $temperature` lowers to a computed eval-output slot, and a reachability marker compiled into that arm was hit ZERO times across all 326 corpus models -- so it cannot be covered by a runtime test and the suite does not claim to. Regression 237/237. Two example suites, one new model.

E-307 *openvaf/sim_back/src/topology/linearize.rs*

A **ddt** with no contributions crashed the compiler. Found by grammar-based fuzzing aimed at the MIDDLE/BACK end (the parser was already hardened; this generator emits well-typed Verilog-A that compiles, so it reaches MIR/optimizer/autodiff/codegen), run against the assertions-enabled build. `linearize.rs` assumed an analog operator reaching the linearizer with an empty contribution list could only be noise: `assert!(noise, "ddt should have been deadcode eliminated")`. False -- a `ddt` whose result never reaches a contribution survives DCE. And it was a PLAIN `assert!`, not `debug_assert!`, so the SHIPPED release crashed ("OpenVAF encountered a problem and has crashed!") on valid input. 5 independent seeds of 3000 hit the identical assert; delta-debug + ablation pinned the trigger to `ddt` + a current probe on a declared (probe-only) branch + `if/else` + `case`, in a module contributing nothing -- removing any one stops it. Fix: `return Evaluation::Dead` unconditionally (the branch already taken for noise; its consumer replaces the result with zero and retargets pending uses), so a `ddt` that feeds no device equation contributes zero. Verify (examples/vafdeadop): reproducer compiles + `.osdi` loads + a CONTRIBUTING `ddt` still gives

E-308 *openvaf/mir_llvm/src/builder.rs*

Uninitialized read feeding a loop-carried phi crashed codegen. Second bug from the same grammar-based middle/back-end fuzz campaign as E-307 (seed 3230 of 8000). A variable read BEFORE a loop that is its only writer leaves the loop-carried phi with an incoming value no reachable block defines: MIR shows `v29 = phi [v18, block7], ...` where `v18`'s defining phi was dropped by a pass on the dead path but the edge kept. Codegen's phi-completion loop then hit `BuilderVal::get()` on a still-`Undef` value -> `unreachable!("attempted to read undefined value")` at `builder.rs:143` -- a plain unreachable, so the SHIPPED build crashed on valid Verilog-A. The MIR verifier misses it (its phi check only tests dominance when the def has a block; a detached def passes; and it is `debug_assert!` anyway). The trigger needs the module to contribute nothing keeping the value live (adding `I(p,n) <+ ra` fixes it). Fix, PROVABLY correct: `build_func` builds every reachable block before completing phis, so a phi input still `Undef` names a value NO reachable block defines (a dead path) -> lower it to `cx.const_undef` of the phi's type rather than panicking. Whole-class fix (any optimizer leaving a value that flows only into dead phi edges), not a per-pass chase. Verify (examples/vafuninitloop): reproducer compiles + a LIVE loop-carried accumulator still reads back exactly `N*g` (proves undef touches only dead inputs); fails on the pre-fix compiler; 328-model corpus identical pass/fail old vs new; 8000-seed re-fuzz shows 0 of this and the E-307 crash. THIRD, rarer pre-existing ICE surfaced and documented not fixed: `packed_option.rs:60 PackedOption::unwrap()` on `None`, ~1/8000, old compiler crashes too.

E-309 *openvaf/mir_opt/src/global_value_numbering.rs*

GVN crashed on a user instruction in an unreachable block. Third and final crash from the same grammar-based middle/back-end fuzz campaign as E-307/E-308 (seed 6716, ~1/8000). When an instruction's congruence class changes, GVN re-queues its USERS via `inst_to_dfs[user].unwrap_unchecked()`. `DFSMapping::populate` numbers only instructions reachable through `cfg_postorder`, so a user in an UNREACHABLE block has no DFS id. `unwrap_unchecked = if cfg!(debug_assertions) { self.unwrap() } else { self.0 }`, so it panicked at `packed_option.rs:60` under debug-assertions and in release returned the reserved sentinel that touched `insts`.

insert used as an OUT-OF-RANGE BitSet index -- the SHIPPED compiler crashed either way ("OpenVAF encountered a problem and has crashed!"). Fix: skip users with no DFS id -- an unnumbered user is in an unreachable block, not in the GVN work list (the solver only iterates `dfs_to_inst`), so re-queuing it is a no-op -- exactly as `get_rank` in the same file already tolerates the identical `None` (if `let Some(dfs_id) = inst_to_dfs[inst].expand()`). The two other `unwrap_unchecked` sites operate on the current instruction (always numbered) and are unchanged. Verify (examples/vafgvnunreach): reproducer compiles + a CSE-heavy model GVN actively optimises still computes exact $I = 4Vg + (V * g)^2$ (proves the pass is undisturbed); fails on the pre-fix compiler; 330-model corpus identical pass/fail old vs new. A 12000-seed re-fuzz against the fully-fixed compiler shows 0 of this and the E-307/E-308 crashes -- the three distinct crashes this campaign found are all closed. That deeper run also tripped `ONE debug_assert!(cx.func.validate())` at `sim_back/src/lib.rs:175` (seed 11633): NOT a shipped crash (release compiles it fine), but the same assertions-only malformed-MIR class as E-286..E-294, documented for a separate follow-up.

E-310 `openvaf/mir_opt/src/simplify_cfg.rs`

Constant-branch fold left an SSA-invalid phi (the `sim_back/lib.rs` validate assert). The MIR-validity defect the E-307/308/309 fuzz campaign surfaced, now RESOLVED. `const_fold_terminator` folds `'br TRUE`

E-313 `openvaf/hir_ty/src/inference.rs`

Two builtin argument type-coercion gaps, both emitted silently by the release compiler. Found by a fresh round of the same grammar-based middle/back-end fuzz campaign as E-307..310, run against the assertions build. (a) FILE/STRING FORMAT TASKS were never type-checked. `infern_display` parses the format string and inserts the `int->real` cast a `%g/%e/%f/%r` conversion needs, but was reached only by the `CONSOLE` tasks (`$display/$strobe/$write/$monitor/$debug + $fatal/$warning/$error/$info`); the `FILE` tasks (`$fdisplay/$fwrite/$fstrobe/$fmonitor/$fddebug`) and `STRING` tasks (`$swrite/$sformat`) were missing from the dispatch match, so their format args were unchecked. A `%g` fed an integer kept its integer value while `print_callback` (`osdi/compilation_unit.rs`) types the callback parameter as `double` from the conversion -- so lowering passed a raw `i32` to a `double` parameter: `INVALID LLVM IR (Call parameter type does not match function signature! i32 3 / double %35 = call ... @cb.2(..., i32 3))`. The verifier that catches this is a `debug_assert!(llmod.verify_and_print())`, compiled out of release, so release emitted a malformed `.osdi` whose callback reads the integer's bit pattern as a `double` -- garbage. Observable: a `$sformat(s,"%g",5) -> $sscanf(s,"%g",g)` round-trip used as a conductance reads back the denormal `2.47e-323` (bits of the integer 5) instead of 5, so a 5V device collapses to `~0`. Console tasks were unaffected (they carry the cast). Fix: add the file/string format builtins to the `infern_display` dispatch arm -- `infern_display` scans for string-LITERAL format strings, so the leading `fd` (integer) or destination (string variable) argument is naturally skipped and the real format string found; the console path's exact logic now applies to every format task. (b) `ddx WITH AN INTEGER ARGUMENT` crashed the compiler. `infern_ddx` requires its first argument (the differentiated value) to be `real` via `self.expect::<false>(expr, None, ty, [Val(Real)])` -- but `expect` records the needed cast on its FIRST parameter, and it was handed `expr` (the whole `ddx` call) while `ty` is the type of `val` (the first argument). For an integer `val`, `expect` recorded an `int->real` cast ON THE `ddx CALL` expression, which already has type `Real`; needs_cast then computed `src=Real` (the call's type) and `dst=Real` (the recorded cast) and tripped `debug_assert_ne!(src, dst, \"cast types must be different\")` at `hir/src/body.rs` -- with the assert compiled out, the release build aborted downstream with no `.osdi` (`ddx(n, V(b))`, integer `n`). Fix: record the requirement on `val`, the argument, not on `expr`; an integer argument is then coerced to `real` (`ddx` of a probe-independent value is 0) and a `real` argument is unchanged. Both fixes only touch previously-crashing/invalid paths: the whole 419-model corpus produces BYTE-IDENTICAL MIR before and after (deterministic --dump-mir oracle, 0/419 changed), the corpus replays clean through the assertions build, and a 15000-module re-fuzz on the fixed compiler is clean. Verify (examples/vafargcoerce): 4 checks under both solvers, all failing on the pre-fix binary -- `ddx(integer)`

compiles + simulates to $I=1e-3V$, and `$sformat("%g",integer)` compiles + the round-tripped value is exactly 5. DEFERRED (same campaign, needs a design decision): a provably-infinite analog loop (`while (1) ...`) crashes the compiler via a degenerate no-reachable-exit CFG -- a DCE guard stops the first unwrap but the crash resurfaces in the CFG-validity machinery; left for a dedicated change (reject non-terminating analog loops with a diagnostic, or make the passes tolerate an unreachable exit).

E-314 *openvaf/hir/src/elaborate.rs, openvaf/mir_opt/src/const_eval.rs, openvaf/hir_ty/src/inference.rs*

Constant-evaluation / literal-materialization robustness -- one shipped DoS, two overflow aborts. From the E-307..313 grammar-fuzz family. (a) INTEGER CONST-FOLD OVERFLOW (three sites, one class): two hand-rolled integer const evaluators used UNCHECKED i32 arithmetic. *elaborate.rs's* Enhancement-91 bus-width folder had `checked_mul` but its `parse_add +/- (acc += / acc -=)` and `parse_unary negate (-parse_unary)` were unchecked, so `localparam integer k = 2147483647 + 1;` (with any `[...]` declaration present -- the folder is gated on the module text containing `'[']`) overflowed; and *mir_opt/const_eval.rs*, where Enhancement-286 made `Iadd/Isb/Imul` wrapping and noted `'eval_unary already used this convention'` but MISSED `Opcode::Ineg`, so negating `i32::MIN` (from `-(1<<31)`) overflowed. All three aborted the overflow-checked build; the shipped release wrapped silently. Fix: *elaborate.rs* uses `checked_add/checked_sub/checked_neg` -- declining the fold on overflow exactly as its `*` already did, and `fold_parameter_widths` handles `None` by leaving the declaration unchanged; *const_eval.rs* uses `val.wrapping_neg()`, matching its own wrapping binary ops. (b) UNBOUNDED REPLICATION -> SHIPPED COMPILE-TIME DoS: `{N{...}}` materializes `N` copies at COMPILE time (`infe_concat/lower_string_concat` build an `N*`

E-317 *openvaf/mir_llvm/src/builder.rs*

idt initial-condition in a dead branch crashed codegen. From the E-307..314 analog-operator fuzz family. An *idt(IC)* or *idtmod* placed inside a statically-false branch -- `if (ceil(0) > 1) w = idt(V(a), 0);` -- crashed the shipped compiler (exit 101). *ceil(0)>1* is always false but *ceil()* is NOT const-folded, so the dead branch survives into MIR; because *w's idt initial-condition state* is never used, codegen prunes the branch *CONDITION's* computation as dead, yet the Branch instruction survives into the derived *osdi::setup::setup_instance* function. When *build_func* reaches that branch and reads its condition, the condition's *BuilderVal* is still *Undef*, and *BuilderVal::get* hit *unreachable!("attempted to read undefined value")* (*mir_llvm/builder.rs:143*). Only *idt/idtmod* (integrator accumulator STATE with an IC) trigger it -- *ddt/absdelay/transition/slew/laplace*/bare idt(x)* are clean. Same *Undef-value* class as Enhancement-308 (which handled the PHI-INPUT case); this is the BRANCH-CONDITION case. Fix: when a Branch's condition is *Undef*, lower it as constant false -- the branch only guards dead code (the unused *idt-IC* state *init*), so the guarded path never executes on either edge, making this observationally equivalent and avoiding an undefined br. Verified corpus-bit-identical (only the reproducer, which was crashing, changes; the other 419 models are byte-identical MIR). Verify (examples/vafidtcfg): reproducer compiles (crashed before) + simulates to a finite op (*i(v1)=-5e-4*). DEFERRED from the same hunt: an assertions-only PHI-type mismatch (an uninitialized integer variable's default materialised as *f64 F_ZERO* -> *i32-result phi* with an *f64* operand; release folds both to 0, correct) and a *ngspice .tran* convergence livelock on a specific fuzzer OSDI device (tiny *1e-15 cap + clamped exp diode + tanh + noise*) -- both left for dedicated changes.

E-324 *openvaf/hir_lower/src/expr.rs*

\$fatal stranded code in an unreachable block -- TWO shipped-compiler crashes, one root cause. Found by a diverse-strategy fuzz campaign against the assertions build (~75000 generated well-typed models, 7 generation strategies); both crashes reproduce on the SHIPPED release, not just the instrumented build. `analog begin $fatal(0); I(a,c) <+ V(a,c)/1k; end` panicked at *mir_opt/src/dead_code_aggressive.rs:112* (`Option::unwrap()` on `None` from `self.func.layout.inst_block(inst).unwrap()` in `mark_inst_live`), and the mirror image `analog`

begin I(a,c) <+ V(a,c)/1k; \$fatal(0); end panicked at mir_llvm/src/builder.rs:143 (unreachable!("attempted to read undefined value") in BuilderVal::get, reached from store_residual). ROOT: \$fatal lowered to print -> set the abort flag -> exit() -> create a fresh block with NO incoming edges -> keep lowering into it, on the assumption (stated in its own comment) that an edge-less block is simply removed from MIR. That is unsound for a compiled device: every ret-flag (RetFlag::Abort/Finish/Stop) is only a flag the simulator inspects AFTER the eval function returns -- all three lower to set_ret_flag_* callbacks in osdi/src/compilation_unit.rs -- so none can jump out of the middle of an evaluation, and the OSDI eval function has a MANDATORY epilogue (store residual/jacobian) the ABI requires to run. Terminating the MIR function early therefore stranded work in the predecessor-less block: a statement AFTER \$fatal was lowered there yet stayed referenced by the contribution bookkeeping (values held OUTSIDE the DFG are not reached by CFG cleanup -- the E-294 class), so aggressive DCE marked live an instruction belonging to no block; and with a statement BEFORE it the EPILOGUE ITSELF was emitted there, where the residual value does not dominate, so codegen read an Undef. \$finish and \$stop were already lowered correctly as set-flag-and-continue (no exit(), no unreachable block) -- \$fatal was the lone outlier. FIX: \$fatal sets its flag and falls through exactly like \$finish/\$stop; the CFG stays connected so neither failure mode can arise (one hunk, one file). Measured trigger surface: crashes required an UNCONDITIONAL \$fatal plus a contribution in the module -- a conditional if(..) \$fatal(0); never crashed (the join block is still reachable via the else path), nor did \$fatal with no contribution, nor \$finish/\$stop in any position. BEHAVIOUR UNCHANGED: \$fatal's run-time meaning lives entirely in the set_ret_flag_fatal callback, untouched -- simctrl_examples (E-55) passes in full including "message printed", "abort error printed", "transient aborted early" and "parameter-only fatal : setuprejected"; vafcodegen_examples(E-294)19/19includingtheparameter-guarded`fatal arm which still SIMULATES correctly (I=V/1k); and a realistic guard (if(r<=0) fatal(...)`followedbya`V/r`divide)stillaborts cleanlyat setupwith its message and no NaN -- statements after`fatal now execute instead of being dead -- is exactly what`finish`/`stop have always done and is not observable, since the simulator aborts as soon as eval returns. OUTPUT-PRESERVING:--dump-mir(deterministic, unlike .osdi bytes) over the full 462-model corpus is BYTE-IDENTICAL for 460; the only two that differ are precisely the models using fatal in areachable analog context(simctrl_demo.va, staleuse.va), both of whose suites pass in full are unchanged. Verify (examples/vaffatalcfg): the two crash shapes compile (both fail on the pre-fix binary with a compiler panic and no .osdi) plus run-time behaviour guards; 6 checks.

E-325 `openvaf/hir_ty/src/inference.rs`, `openvaf/hir_ty/src/diagnostics.rs`

Bound the MATERIALIZED SIZE of `{...}/{n{...}}`, not just the replication count -- a shipped compile-time HANG and a shipped silent WRAP. Enhancement-314 capped the replication COUNT at 2^{20} after a `{'d999999999{"x"}}` DoS, but the count is only ONE FACTOR of the final size, so two abusive shapes still reached the shipped compiler (found by the 7-strategy ~75000-model fuzz campaign). (a) STRING replication becomes LLVM FUNCTION ARITY: lower_string_concat builds an `elems.len()*rep_cnt` operand list plus a format string with that many `%s`, and `print_callback` turns it into a generated callback with ONE PARAMETER PER OPERAND. LLVM degrades super-linearly in arity -- measured on this compiler: 2000 operands 0.4s, 8000 2.9s, 16000 8.6s, 32000 never finished -- so parameter string `s = {200000{"x"}};` HUNG the compiler on one line of source (both binaries). NOTE the original hypothesis (quadratic string interning in the const-folder) was WRONG: the const-fold is linear and negligible; the entire cost is in LLVM. (b) The size arithmetic OVERFLOWED u32: `len: total * rep_cnt` was an unchecked multiply (and the running `total += len` equally unchecked), so `real c[0:1]; c = {1048576{{1048576{1.0}}}}`; -- where BOTH counts are individually legal (exactly MAX_REP) -- computed $2^{20} * 2^{20} = 2^{40}$, which panicked under overflow-checks and in the SHIPPED release silently WRAPPED TO 0, emitting the nonsense diagnostic expected `real[0:2]` value but found `real[0:0]` value instead of a real error. FIX: compute the

size in u64 with saturating add/mul and bound it BEFORE narrowing to the u32 array length, with a dedicated ConcatTooLarge diagnostic that reports the TRUE expanded size (expands to 1099511627776 elements = 2^{40} , where the u32 path had wrapped to zero). Reusing the existing InvalidReplicationCount diagnostic would have been MISLEADING -- in case (b) the count IS a valid positive integer literal; it is the product that is too large. TWO limits, because the two paths have genuinely different measured cost profiles: MAX_CONCAT_ELEMS = 2^{20} for numeric arrays (that path is linear and cheap -- 65536 elements compile in 0.41s -- so this only rules out the absurd and guards the u32 length) and MAX_CONCAT_STR_OPERANDS = 4096 for strings (arity is the real cost; 4096 keeps the worst case near a second, orders of magnitude above any legitimate literal). LRM POSITION: $\{n\{\dots\}\}$ over strings/1-D arrays is this project's Enhancement-34 EXTENSION -- Verilog-A (LRM Annex C) has no vector concatenation at all and Verilog-AMS defines $\{ \} / \{n\{ \} \}$ over bit vectors -- so no LRM-mandated semantics is violated by a documented expansion limit (the precedent E-314 set); and for (b) any expansion larger than the destination is a TYPE ERROR today, the only defect being that the compiler computed the size wrongly before it could say so. No legal program changes meaning. OUTPUT-PRESERVING: u64 saturating add/mul agree exactly with the old u32 wrapping arithmetic for every value below 2^{32} and both caps are checked before the narrowing, so every model whose concatenation fits takes a bit-identical path -- confirmed with the deterministic `--dump-mir` oracle over the 465-model corpus: 464 byte-identical, and the single reported diff (lrm_p150_1.va, which contains NO concatenation at all) was PROVEN to be run-to-run nondeterminism of the multi-module dump order, not an effect of the change (the same binary on the same input produced both hashes across 5 runs). Verify (examples/vafconcatsize): the three abusive shapes are rejected cleanly (they hang or misreport on the pre-fix binary), the diagnostic reports the true 2^{40} size, and legitimate concatenation still compiles AND simulates (a replicated array element drives $i(v1) = -2$ mA exactly); 6 checks.

E-326 *openvaf/sim_back/src/init.rs*

A cross-namespace `mir::Value` comparison mis-typed init cache slots -- the shipped compiler SIGSEGV'd inside LLVM. Highest-severity finding of the 7-strategy ~75000-model fuzz campaign: on a legal Verilog-A model the RELEASE binary died with EXC_BAD_ACCESS in `llvm::detail::DoubleAPFloat::multiply` (signal 11, no diagnostic); the assertions build instead had its LLVM verifier reject the module with `fmul reassoc .. i8 %8, double %6 and trunc double %45 to i8`, and at -O0 release gave LLVM ERROR: Do not know how to promote this operator!. A boolean-typed (i8) value was reaching floating-point arithmetic. ROOT: NOT a missing cast, NOT autodiff, NOT casex. A `mir::Value` is a bare u32 index and the MAIN function and the INIT function each number their values from zero. `build_init_tern` inserts `self.val_map[&val]` -- an INIT-namespace value -- into `collapse_implicit`, which is correct for its other consumer `optimize()` (that runs DCE over `self.init.func`); but `build_init_cache` iterates `self.init_cache`, whose keys are MAIN-namespace values, and tested them against that same set (`collapse_implicit.contains(&val)`) both in its dead-value filter and in its type selection. Comparing indices across two namespaces cannot fail loudly -- it silently SUCCEEDS whenever the two counters coincide. Confirmed under lldb on the minimized reproducer: the producer inserted `init Value(25)` (the `idt_collapse` flag, a genuine Bool) while the consumer tested `main Value(25)`, which is `abs(p)` -- the NOISE POWER, a Real. Same number, unrelated values. CONSEQUENCE: a colliding f64 slot was recorded as `Type::Bool`, which lowers to `ty_c_bool() = i8`. The store side (`store_cache_slot`) then int-cast a slot it believed boolean, emitting `trunc double .. to i8`; and the noise loader reads `EvalOutput::Cache RAW` with the slot type -- unlike `load_cache_slot` it performs no bool normalization -- so the i8 became `base_pwr` and landed in `fmul i8 %x, double %y`. LLVM constant-folded that malformed `fmul` and read the i8 as an APFloat: the EXC_BAD_ACCESS. FIX: map through `val_map` before the lookup so both sides speak the same namespace -- `let is_collapse_flag = self.val_map.get(&val).map_or(false,`

E-327 *openvaf/hir_lower/src/{expr.rs,ctx.rs,lib.rs}*

ddx unknowns that do not lower to a bare MIR **Param** crashed the shipped compiler -- the bug was sitting behind a **TODO**. From the 7-strategy fuzz campaign. **ddx** lowering assumed its UNKNOWN argument always lowers to a bare Param and unwrapped one unconditionally: the DDX_POT arm did `let node = self.ctx.unwrap_node(unknown); (itself value_def(val).unwrap_param())` directly under a `// TODO how to handle gnd nodes? comment`, and the other arm did `CallBackKind::Derivative(self.ctx.dfg().value_def(unknown).unwrap_param())`. That assumption is false: `LoweringCtx::nodes` yields THREE shapes -- a Param for a forward-oriented probe $V(a, b)$, $V(a)$; an **fneg(param)** INSTRUCTION for a REVERSE-oriented probe $V(b, a)$, or one whose high side is ground, via either the `(None, Some(lo))` arm or the inverted-param arm); and **F_ZERO**, a constant, when the probe is ground only. The latter two hit `unwrap_param()` and panicked `Value is not a parameter (mir/src/dfg/values.rs)` on BOTH binaries -- a shipped crash on legal input. DECISION -- COMPILE, do not reject: both extra shapes have an unambiguous derivative. $V(b, a)$ denotes the SAME branch with the opposite reference direction, i.e. $V(b, a) == -V(a, b)$, so $df/dV(b, a) == -(df/dV(a, b))$; and ground is not an unknown of the DAE system, so $df/dV(gnd) == 0$ -- exactly what the backend already does for a derivative callback that never became an unknown. Erroring would also be incoherent: `openvaf` already ACCEPTS $V(a, b)$ as a **ddx** unknown (an extension beyond the LRM's single-net/branch-flow rule, announced by its own L011 lint), so having accepted $V(a, b)$ it cannot reject $V(b, a)$. FIX: peel an **fneg** off the unknown (recording that the result must be negated), then ASK rather than assert whether what remains is a parameter -- `match self.ctx.dfg().value_def(probe).as_param() { None => F_ZERO, Some(param) => ... }` -- negating the callback's result when an **fneg** was peeled; plus a non-panicking `ParamKind::pot_node()` replacing `unwrap_pot_node()` on the DDX_POT path and a `LoweringCtx::param_kind()` accessor so lowering can inspect a user-supplied unknown instead of asserting its shape. VERIFIED NUMERICALLY, not merely that it compiles: differentiating V^2 (exact derivative $2V$) at $V=3$ through a 1 mS scale gives $i = -6.00000e-03$ for the forward unknown $V(a, b)$, $i = +6.00000e-03$ for the reverse unknown $V(b, a)$ -- the EXACT negative to machine precision -- and $-3.00000e-09$ for the ground unknown, which is purely the model's explicit $1e-9 \cdot V$ leak term, i.e. the **ddx** contributed exactly zero. OUTPUT-PRESERVING: the new code diverges from the old only when `value_def(unknown)` is NOT a Param, which is precisely the case that previously reached `unwrap_param()/unwrap_node()` and panicked, producing no output at all; when the unknown IS a Param (every model that compiles today) the **fneg** peel does not fire and the callback construction is unchanged. Confirmed with the deterministic `--dump-mir` oracle over the corpus: 0 changed. Verify (examples/vafddxunknown): the two crashing shapes compile and their derivatives are checked against the closed form; 4 checks.

E-328 `openvaf/hir/src/body.rs`, `openvaf/hir_lower/src/expr.rs`

BodyRef::get_expr was a PARTIAL function used as if it were total -- a dynamic array index in a contribution crashed the shipped compiler. `get_expr` funnelled every `BitSelect` into `resolve_path`, which can only resolve expressions that have a `Ref (Ty::Var, Ty::Param, a function var/return, a nature attribute)` and `panic!`s otherwise. A dynamically-indexed array read has NO backing variable: inference types it `Ty::Val(..)` and records the element variables, per-dimension bounds and index expressions OUT-OF-BAND in `dynamic_index_refs`. So any caller that merely probed an expression's SHAPE -- a literal-zero test, a literal-condition fold, an aggregate check -- crashed on legal input with invalid HIR: `path BitSelect { .. }` was not resolved `Val(Real)`. The asymmetry is the tell: `x = g[k]; I <+ V*x;` compiled while `I <+ V*g[k];` panicked -- `lower_expr` short-circuits on `dynamic_index()` BEFORE consulting `get_expr`, and the contribution path has no such short-circuit. (Note an earlier analysis reported this shape as already working; re-verification on the shipped binary showed it still crashed in both the uninitialised-index and assigned-index spellings -- worth re-testing rather than trusting.) FIX: restore the invariant that `get_expr` is TOTAL -- add an `Expr::DynIndexRead` variant and answer the shape question directly, gated on the same `dynamic_index_refs` map the value path already consults. The VALUE is still lowered by `lower_expr`'s existing `dynamic_index()` short-circuit,

so nothing about code generation changes. Making the enum non-exhaustive pointed the compiler at EXACTLY ONE match needing an update (lower_expr's, which already short-circuits and so gets an unreachable! arm), confirming how tightly scoped the change is. VERIFIED NUMERICALLY: a dynamic index must select the RIGHT element, not merely stop crashing -- four instances selecting $g = \{1,2,3,4\}$ mS at $V=1$ give exactly $-1.00000e-03/-2.00000e-03/-3.00000e-03/-4.00000e-03$, agreeing bit-for-bit with the $x = g[k]$ spelling that always worked. OUTPUT-PRESERVING: the new branch is gated on `dynamic_index_refs.contains_key(&expr)`; for any expression not in that map -- every expression in every model that compiles today -- `get_expr` executes exactly the previous code path, and for an expression that IS in it the previous behaviour was a panic, so there is nothing to preserve. Confirmed with the deterministic `--dump-mir` oracle. Verify (examples/vafdynidx): 3 checks.

E-329 *openvaf/sim_back/src/topology/small_signal_network.rs*

A GRAVESTONE phi operand crashed the small-signal network builder. ddt of a NEGATED flow probe plus a statically-false branch containing a loop -- `r0 = ddt(-I(a,b)); if ((1*s0)-s0) begin lc3=0; while (lc3<4) lc3=lc3+1; end I(b,c) <+ r0*r1;` -- panicked the SHIPPED compiler with internal error: entered unreachable code. ROOT: the value arriving is GRAVESTONE, `Value(0)`, whose `ValueDef` is `Invalid`. It is the compiler's OWN placeholder, declared in `mir/src/dfg/values.rs` as "place holder for unused values that must remain (in phis)": the SSA re-builder puts it in a phi for an edge with NO reaching definition -- an edge from a block unreachable from the entry that `simplify_cfg` cannot delete. The small-signal network builder asserted such a value could never reach it, with `ValueDef::Invalid => unreachable!( )` in BOTH `analyze_value` and `analyze_dependency`. FIX: a GRAVESTONE operand sits on a dead edge and therefore cannot be used at run time, so it contributes nothing -- `analyze_value` returns `FlatSet::Zero` (the same answer its neighbouring `F_ZERO` arm gives) and `analyze_dependency` returns `Dependency::Independent` (the same answer its `Param( )`

E-330 *openvaf/hir_ty/src/validation/body.rs*

ddx in a runtime loop HUNG the shipped compiler forever -- a fixpoint that grows its own lattice. The last of the seven shipped crashes from the diverse-strategy fuzz campaign, and the only infinite loop rather than a panic: `for (i=0;i<1;i=i+1) x = ddx(V(a)*x, V(a));` never returned on either binary. ROOT: `live_derivative_fixpoint` (`mir_autodiff/src/live_derivatives.rs`) asks, for every derivative already live at a `ddx` call, for one of `ORDER+1` via `raise_order_with`. That is fine on a DAG -- the order chain is bounded by the number of `ddx` sites on the longest path. A loop back edge closes the circuit: the differentiated expression `V(a)*x` depends on `x`, which is the loop-carried phi fed by the `ddx` result itself, so round `n` creates order `n+1` and interns it, `populate_reachable` marks it reachable at that instruction, the changed live set re-queues the argument's defining instructions, and the back edge carries the new order around the phi back into the `ddx`'s own input -- round `n+1` begins. A monotone fixpoint terminates because its lattice is FIXED; here the fixpoint GROWS THE VERY LATTICE IT ITERATES OVER, so it has no fixed point. Measured not inferred: `sample` puts 99.8% of 3944 samples in that one chain (`raise_order_with -> TiSet::ensure -> IndexMap::insert_full`), RSS climbs monotonically and never plateaus, and the process was still running after 15 minutes -- true non-termination, not slowness. DECISION -- clean error, not a cap: `ddx` is SYMBOLIC and MEMORYLESS (`openvaf` computes it as a derivative callback on the MIR; its result is determined by the SHAPE of its argument). Inside a runtime loop `x = ddx(V(a)*x, V(a))` asks for a different symbolic form every trip -- iteration `k` needs the `k`-th derivative and the trip count is not a compile-time constant -- so THERE IS NO FINITE MIR THAT IMPLEMENTS IT, which is exactly why the fixpoint diverges rather than converging to something wrong. Ill-formed, not legal-but-unbounded. This is also what the language already says and what the compiler already does for every OTHER analog operator: `ddx` is classified as an analog operator by `openvaf` itself (`hir_def/src/builtin.rs` is `_analog_operator()`) and LRM 4.5.1 forbids analog operators in non-genvar loops -- `for(...)` `x = ddt(V(a));` and `for(...)` `x = idt(V(a)*x);` already produce analog operator ... is not allowed in

loops. The divergence came from one explicit escape hatch in `hir_ty/src/validation/body.rs`, `if call.is_analog_operator() && call != BuiltIn::ddx`, which is CORRECT for conditionals (the industry CMC corpus has 192 ddx call sites inside if bodies across 19 models -- BSIM-BULK, BSIM-SOI, HICUM, ASM-HEMT, L-UTSOI) but too broad, since it also covers loops. FIX: track runtime-loop nesting in a dedicated `loop_depth` counter and reject ddx there, reusing the existing `IllegalCtxAccess` diagnostic so it is an ordinary compile error (exit 65, no new machinery). A separate counter is required rather than reusing `ctx`: `validate_condition_in` REPLACES the context instead of stacking it, so an if nested in a for resets it to Conditional, and it only becomes Loop when the controlling expression is non-constant, so `repeat(3)` would slip through. Trigger surface verified: for, while and repeat all hung, as did a ddx under an intervening if and one routed through a second variable; the multiplication is NOT essential (`ddx(V(a)+x, V(a))` hangs too) -- the precise condition is that the argument depends on the differentiation unknown AND on the ddx call's own result through a back edge. SCOPE, STATED PLAINLY: ddx outside a loop is untouched, including inside if/else, and is still numerically exact there ($d/dV(V^2)=2V$ gives -6 mA at $V=3$). A corpus scan of 514 models finds 0 of 755 ddx call sites inside a loop body, and the -dump-mir oracle reports no corpus model changed. This is nonetheless the ONE fix in this series that slightly NARROWS the accepted language: a loop containing a ddx with no self-reference (`for(...) g = g + ddx(V(a)*V(a), V(a));`) compiles today and now errors. It is absent from the corpus, the rejection is LRM-conformant, and it is exactly the treatment `ddt/idt/transition/laplace_*` already receive -- but it is not strictly output-preserving and is not claimed to be; keeping it would require front-end dataflow that can distinguish self-referential from not, which the validator does not have. SEPARATELY DISCOVERED, NOT FIXED HERE: while bounding the derivative order a SECOND shipped crash surfaced -- 65 nested ddx calls panic in `lib/bitset/src/lib.rs` (`index out of bounds: the len is 1 but the index is 1`) via `HybridBitSet::contains`, reached from `populate_reachable`; a row that has gone dense keeps the word count it had at that moment, so once the derivative universe grows past 64 (one word) a query on the new index panics. Independent of loops and of this fix, left for a dedicated change. Verify (examples/vafddxloop): the hanging shape is now a prompt error citing the LRM, and ddx outside a loop still compiles AND stays exact; 4 checks.

E-331 *lib/bitset/src/lib.rs*

BitSet::contains panicked outside its domain -- a representation leaking out as a compiler crash. Found while bounding the derivative order for E-330: deeply nested ddx crashed the SHIPPED compiler with `index out of bounds: the len is 1 but the index is 1` at `lib/bitset/src/lib.rs:114`, via `BitSet::contains` `<- HybridBitSet::contains` `<- mir_autodiff::populate_reachable`. ROOT: `contains` indexed its backing storage with no bounds guard -- `(self.words[word_index] & mask) != 0`. A `HybridBitSet` stores a set either SPARSELY (a `Vec` of elements) or DENSELY (a bit array), switching as a size optimisation, and rows grow their domain only when something is inserted into them -- so a row the traversal never inserts into keeps whatever width it had when it went dense. Once the live-derivative universe grew past 64, i.e. ONE WORD, a query for a higher element indexed off the end. The decisive detail is the ASYMMETRY between the two representations: `SparseBitSet::contains` searches a `Vec` and returns false for any absent element, while the dense `BitSet::contains` panics -- so the SAME logical query on the SAME logical set returned false while the set happened to be sparse and CRASHED once it happened to be dense. The representation, the very detail `HybridBitSet` exists to hide, leaked out as a crash. FIX: make the dense query total, matching what the sparse one already does -- `'self.words.get(word_index).map_or(false,`

E-332 *openvaf/sim_back/src/topology/builder.rs, openvaf/hir_ty/src/validation/body.rs*

Summing THREE OR MORE **ddt()** terms into one contribution silently DROPPED CHARGE -- a wrong number, not a crash. `I(a,b) <- ddt(V)+ddt(V)+ddt(V)` compiled cleanly and behaved as a 1 F capacitor instead of 3 F; $N=1$ and $N=2$ were correct, which is why it survived. Reachable from entirely ordinary code: a `generate` loop is unrolled at compile time, so `generate k (0,2)`

`s = s + ddt(V);` is the same expression and was equally wrong. ROOT CAUSE was NOT in `ddt` handling but in TRAVERSAL ORDER. `create_dimension` replays the instructions consuming an analog operator's result to build its linear coefficient, and its Fadd/Fsub arms '(None, Some(x))